THE TERMINAL MEASURE AND PRESSURE ESTIMATES FOR LATTICE QED IN THREE DIMENSIONS

Abstract. We identify the terminal measure in the block construction of three-dimensional lattice QED with chamber fermions and prove a common-pressure estimate on its prescribed family of tori. The measure identity holds for arbitrary bounded Borel observables and retains the normalization of every restricted Gaussian integral. A zero-flux Gaussian bound controls the large-field complement. A mesh-uniform determinant estimate then yields a partition comparison for two reference tori, with a bound independent of the ultraviolet cutoff.

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1. Introduction

We use the chamber regulator with weights $(6, 6\sqrt{2}, 6\sqrt{2}, 6)$ and Wilson parameter $0 < \rho < 1$. On the unit cubic torus C_n with n sites per side, let $\mathcal{F}_n = dC^1(C_n; \mathbb{R})$ be the space of exact real two-cochains, with counting inner product and standard Gaussian measure γ_n. Write $A(u)$ for the mean-zero coclosed potential of $u \in \mathcal{F}_n$. For $q > 0$ and a real lattice mass $\boldsymbol{\mu}$, define the matter factor

$$z_{r,n}(q, \boldsymbol{\mu}) = \int_{[-\pi,\pi]^3} \frac{\mathrm{d}\theta}{(2\pi)^3} \int_{\mathcal{F}_n} \det\bigl(D^{\mathrm{ch}}[q(A(u) + H_\theta)] + \boldsymbol{\mu}\bigr)\, \mathrm{d}\gamma_n(u), \qquad H_{\theta,\nu} = \frac{\theta_\nu}{qn}. \tag{1}$$

Here $D^{\mathrm{ch}}[qB]$ is the unit-mesh, four-spin-component chamber Dirac operator of [26f, §2], with 48 chamber labels per site and antiperiodic spinors. Its argument qB is the bond phase; the subscript r in $z_{r,n}$ denotes the regulator. The harmonic average in (1) is the zero-flux matter factor of [26c, (37)–(40)].

Fix once and for all a choice of the block construction constants satisfying [26h, Proposition 4.1], through its charge ceilings $e_1 \le e_0 < 1$ and terminal scalar conditions. Thus L is odd, M is a power of L, $M \ge L^2$, $0 < \varepsilon \le 1/200$, and the exponents are $r = 2$, $p_0 = 9$, $p = 19$. The fixed ceilings denoted by e_0, e_1 here are the $\bar{e}_0, \bar{e}_1$ of [26h]; they are distinct from the running charge e_j at $j = 0$. This choice includes the photon and fermion primitive bounds, the exact normalized block step, and its field domains. For $0 < \bar{m} \le 1$ and $N \in \mathbb{Z}_{\ge 0}$ put

$$h = L^{\lfloor \log_L \bar{m}^{-1} \rfloor}, \qquad \eta = L^{-N}, \qquad K = N + \log_L h, \qquad a_j = L^{j-N}, \quad e_j = e\sqrt{a_j}, \quad \mu_j = \bar{m} a_j.$$

We distinguish the initial charge $q = e\sqrt{\eta}$ from the fixed ceiling e_0. Set $v_j = -\log e_j$, $p_j = v_j^{19}$, $p_{0,j} = v_j^9$, and let r_j be the least nonnegative integer power of L which is at least v_j^2. Define

$$R_{\mathrm{top}} = hMr_K, \qquad R_* = R_{\mathrm{top}}/L, \qquad \mathcal{T}(e) = \{hL^t : t \in \mathbb{Z}_{\geq 0},\ hL^t \leq R_{\mathrm{top}}\}.$$

For $R \in \mathcal{T}(e)$, the threshold $N_{\min}(R,e)$ and first stopping level $k_*(R) \geq 1$ are those of [26a, Proposition 6.1 and (52)]: the stop is the first level with $\ell_j = Mr_j$, where $\ell_j = R/a_j$. At cutoffs $N \geq N_{\min}(R,e)$, write $a_* = a_{k_*}$, $e_* = e_{k_*}$, $\mu_* = \mu_{k_*}$ and $\ell_* = \ell_{k_*}$. These physical quantities are independent of N, and

$$Mv_*^2 \leq \ell_* < LMv_*^2, \qquad \mu_*^{-1} < L\ell_*, \qquad 0 < e_* \leq e_K = e\sqrt{h}, \qquad 0 < \mu_* \leq 1. \tag{2}$$

The threshold is finite and nonincreasing in R. The numerical consequences of the fixed scheme used below are listed and verified in section 5.

Use the real pair $P_N(e) = (m_{0,N}(e), \varepsilon^{\mathrm{v}}_{0,N}(e))$ of [26a, Theorem D]: it is selected on R_{top} with zero initial activity and zero terminal mass shift and vacuum term there. The same pair gives a controlled trajectory on every $R \in \mathcal{T}(e)$ above its stated threshold. In particular, $|m_{0,N}(e)| \leq \frac{1}{8}q^{3/4-8\varepsilon}$. Set $\boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}(e)$ and abbreviate $z_{r,n} = z_{r,n}(q, \boldsymbol{\mu}_N)$, $n = R/\eta$. The total mass $\boldsymbol{\mu}_N$ may have either sign. Its free comparison uses the positive reference mass:

$$z^{\mathrm{free}}_{r,n} = \int_{[-\pi,\pi]^3} \det(D^{\mathrm{ch}}[qH_\theta] + \bar{m}\eta)\frac{\mathrm{d}\theta}{(2\pi)^3}.$$

Let $\mathcal{D}_m(p)$ denote the 192×192 chamber Fourier symbol at mass $m > 0$, and set

$$p^0_{\mathrm{ch}}(m) = \int_{[-\pi,\pi]^3} \log\det \mathcal{D}_m(p)\frac{\mathrm{d}p}{(2\pi)^3}, \qquad p_F = p^0_{\mathrm{ch}}(\bar{m}\eta) + \varepsilon^{\mathrm{v}}_{0,N}(e). \tag{3}$$

The positive constants $C_6, \kappa_{(0)}$, depending only on ρ and the fixed weights, are given in proposition 4.1. Write $\varepsilon_5(x) = C_6\log(1 - e^{-\kappa_{(0)}x})^{-1}$ and $N_{\mathrm{ref}}(e) = N_{\min}(R_*, e)$.

Theorem A (Common pressure and the two-torus comparison)**.** Fix the construction and the common pair specified above, and assume $0 < e\sqrt{h} \leq e_1$. For every $R \in \mathcal{T}(e)$ and $N \geq N_{\min}(R,e)$, the matter factors $z_{r,n}$ and $z^{\mathrm{free}}_{r,n}$ are positive, and

$$\begin{aligned}\left|\log z_{r,n} - \log z^{\mathrm{free}}_{r,n} - \varepsilon^{\mathrm{v}}_{0,N}(e)n^3\right| &\leq 3e_*^{1/4-9\varepsilon},\\ \left|\log z_{r,n} - p_F n^3\right| &\leq \varepsilon_P(R) := \varepsilon_5(\bar{m}R) + 3e_*^{1/4-9\varepsilon} \leq \varepsilon_5(1/L) + 3e_K^{1/4-9\varepsilon}.\end{aligned} \tag{4}$$

In particular let $R_* = R_{\mathrm{top}}/L$, $n_* = R_*/\eta$, and $N \geq N_{\mathrm{ref}}(e)$. The two tori R_* and LR_* use the same mass and the same pressure. Their matter comparison satisfies

$$\begin{aligned}|D_{\mathrm{mat}}| := \left|L^3\log z_{r,n_*} - \log z_{r,Ln_*}\right| &\leq L^3\varepsilon_P(R_*) + \varepsilon_P(LR_*)\\ &\leq (L^3+1)\left[\varepsilon_5(\bar{m}R_*) + 3e_K^{1/4-9\varepsilon}\right] \leq C_{\mathrm{mat}},\\ C_{\mathrm{mat}} &= (L^3+1)(\varepsilon_5(1/L)+1).\end{aligned} \tag{5}$$

The same $p_F = p_F(N, e, \bar{m})$ is used for every asserted side. The displayed error bounds are uniform in N; C_{mat} depends only on L, ρ and the fixed chamber weights. ■□

The theorem compares volumes within one fixed construction. Its pressure p_F may depend on the cutoff. The estimate does not require a limit of that pressure: the common bulk term cancels exactly between the two reference tori.

The proof is the terminal integration argument for the block construction in [26j, §21–22]. Its block averaging and background-field methods belong to the approach of Bałaban and collaborators; Dimock develops this approach for QED in three dimensions in [Dim22, §1–2]. The chamber determinant, harmonic averaging, and comparison proved here are those of [26j, §22]. We use the previously proved actual block transformation and common tuning in [26d; 26b; 26g; 26i; 26a].

Two estimates connect that transformation to the finite-volume matter factor. First, Gaussian domination in the zero-flux sector bounds the probability of leaving any fluctuation box or the terminal small-curvature set. The sector projection remains inside the trace proof. Second, the exact change from curvature to all photon fluctuation coordinates identifies the restricted measure with a terminal fermion integral. The Gaussian normalization and the restriction to each box are kept in this identity, which holds for every bounded Borel observable.

On the terminal small-field set the interacting integral differs from a positive twisted free determinant by a relative error of order $e_*^{1/4-9\varepsilon}$. The coefficient-norm estimates which prove this are collected in section 6. The free determinant has an explicit bulk pressure; a separate Bloch estimate makes the twist dependence of the block penalty exponentially small in ℓ_*, uniformly in its fine mesh. Combining these bounds with the positive small-field mass gives two-sided partition estimates. Taking their logarithms and bounding the terminal vacuum proves theorem A; the closing argument is in section 5.2.

The generic differentiated block estimates are recorded in appendix A for their use in terminal response estimates. The fixed reference-mass interval used there is separate from the variable range $0 < \mu_* \le 1$ in the pressure proof.

2. Gaussian domination in the zero-flux sector

The complement estimate below concerns the normalized zero-flux law, denoted $\bar{\mu}$: on a cubic torus it is the probability measure obtained by normalizing the integrand of (1) after harmonic averaging. Write F^{u} for the random curvature in unit-lattice coordinates and $F_N = dC^1(C(n_1,n_2,n_3);\mathbb{R})$ for its exact curvature space; on the cubic torus this is $\mathcal{F}_n$. The all-flux Gaussian bound does not directly give a bound for this conditional law. We therefore retain the sector projection in the coordinate-slice trace.

Let $Z^{(0)}[\Phi]$ be the unnormalized zero-flux partition functional with curvature observable Φ, as in [26c, (34)]. Put $r_q = 2\pi/q$ and $g_1(t) = (2\pi)^{-1/2}e^{-t^2/2}$. On $C(n_1,n_2,n_3)$ use the spaces H^s, H^l, P_G, E, and M_0 of [26f, (80)] and [26f, (82)]. Here H^s, H^l are the slice Hilbert spaces at site and link planes, P_G is the gauge-invariant projection, E is the electric momentum operator, and M_0 is the half-slab multiplier including the magnetic Gaussian factor. Let P_k project onto the magnetic-flux summand k. Choose coordinate slice cycles S_2, S_3, through a fixed slice site, and for $\xi \in [0,1]^2$ set $b_\xi = q(\xi_2 \mathbf{1}_{S_2} + \xi_3 \mathbf{1}_{S_3})$. For bounded magnetic factors u_j and electric factors v_j with sub-Gaussian growth, let K_j^ξ have the periodized convolution kernel

$$v_j(A'-A)e^{i\langle b_\xi, A'-A\rangle}\prod_b g_1(A'(b)-A(b)).$$

The zero-flux trace formula is

$$Z^{(0)}\left[\prod_j u_j(F_j^{\mathrm{mag}})v_j(F_j^{\mathrm{el}})\right] = (2\pi)^{n_1n_2n_3}\int_{[0,1]^2}\mathrm{tr}\left(P_0\prod_{j=n_1-1,\dots,0} r_q^{n_2n_3}K_j^\xi P_G M_0 u_j(F^{\mathrm{mag}})M_0^*\right)\mathrm{d}\xi. \tag{6}$$

To derive it, sum the electric curvature along S_2, S_3 and over time. The sums are $r_q k_{12}$ and $r_q k_{13}$. The added electric phases are therefore $e^{2\pi i(\xi_2 k_{12}+\xi_3 k_{13})}$. Restricting the first magnetic slice to flux k_{23} inserts $P_{k_{23}}$ in the word trace, since the sum of $d_s A$ on the slice vanishes. All operators preserve that summand. The flux sum is absolutely convergent by the curvature Gaussian bound on the modulus of the integrand. Set $k_{23} = 0$ and integrate over ξ; Fourier orthogonality keeps precisely $k_{12} = k_{13} = 0$. This proves (6).

Proposition 2.1. For the harmonic-average zero-flux law at every real lattice mass, every coordinate side $n_\mu \ge 4$, and every real two-cochain g,

$$\bar{\mu}(e^{\langle g, F^{\mathrm{u}}\rangle}) \le e^{\|\Pr_{F_N} g\|^2/2}. \tag{7}$$

In particular, for $\phi \in F_N$ with $\|\phi\| \le 1$ and $a \ge 0$, $\bar{\mu}\{|\langle \phi, F^{\mathrm{u}}\rangle| > a\} \le 2e^{-a^2/2}$. ■

Proof. For a real two-cochain β, put $\mathcal{Z}(\beta) = Z^{(0)}[e^{-\langle \beta, F\rangle - \|\beta\|^2/2}]$. It is finite by the all-flux bound and is positive because the sector contains the $z = 0$ term, which has positive integral; the shifted photon weight is strictly positive. For time direction 1, split β into electric parts β_j^{el} and magnetic parts β_j^{mag}. Take the corresponding exponential u_j and v_j in (6), replacing u_j first by $u_j^\delta = \min(u_j, \delta^{-1})$. The bounded magnetic factors make the word trace applicable.

Completing the electric Gaussian gives

$$K_j^\xi = z_j U_j e^{-|E-b_\xi|^2/2}, \qquad U_j = e^{i\langle \beta_j^{\mathrm{el}}, E\rangle}, \qquad |z_j| = 1.$$

Let $K_\xi = r_q^{n_2 n_3} e^{-|E-b_\xi|^2/2} P_G$. It is positive, and its square root is $r_q^{n_2 n_3/2} e^{-|E-b_\xi|^2/4} P_G$. The operator $K_\xi^{1/2} M_0$ is Hilbert–Schmidt. The periodized square-root Gaussian has the additional phase $e^{i\langle b_\xi, A'-A\rangle}$; its modulus is bounded by the kernel majorant used in [26f, (83)]. The same real-slice integral and flux sum therefore proves this assertion uniformly in ξ. Multiplication by bounded $(u_j^\delta)^{1/2}$ preserves it.

Put $Y_j = K_\xi^{1/2} M_0 (u_j^\delta)^{1/2}$. Cyclicity turns the trace into $\prod_j z_j$ times the closed chain

$$\operatorname{tr} \prod_j U_j P_0 Y_j (P_0 Y_j)^*.$$

All factors preserve the magnetic-flux summand; U_j is unitary. Schatten Hölder [Pon25, slides 53–56] with $2n_1$ factors of exponent $2n_1$ bounds this trace by

$$\prod_j \left(\operatorname{tr}_{H^s} P_0 (Y_j^* Y_j)^{n_1} \right)^{1/n_1}.$$

Hölder in ξ and the Bloch trace formula identify its jth integrated factor with the zero-flux functional having the same magnetic factor at every slice and no electric factor. Let $\beta^{(j)}$ denote that purely magnetic two-cochain. Monotone convergence as $\delta \downarrow 0$ gives

$$\mathcal{Z}(\beta) \le \prod_j \mathcal{Z}(\beta^{(j)})^{1/n_1}.$$

Rotate the torus by an orientation-preserving cyclic coordinate permutation. Zero flux and the pairing are preserved, and $\beta^{(j)}$ becomes purely electric for the new time direction. The same inequality then bounds $\mathcal{Z}(\beta^{(j)})$ by $\mathcal{Z}(0)$. Hence $\mathcal{Z}(\beta) \le Z^{(0)}[1]$.

The zero-sector identity [26c, (34)] identifies its normalized curvature law with $\bar{\mu}$. Since this law is supported on F_N, the pairing with g equals that with $\Pr_{F_N} g$. Apply the preceding inequality with the negative of that projection and divide by the positive zero-sector partition function. This proves (7). Exponential Markov applied to the two signs and optimized at parameter a gives the stated tail bound. □

3. The terminal measure

We first put the curvature Gaussian in coordinates adapted to all block steps. The energy splits into the terminal photon energy and independent unit Gaussian fluctuation coordinates. Iterating the exact density identity in these coordinates then gives the terminal fermion integral on the small-field set.

3.1. **The downward photon coordinates.** Fix a flow side and cutoff as in [26a, Proposition 6.1]. Put $k = k_*(R)$ and $n = R/\eta = L^k\ell_*$. On the unit torus $T_j^0 = \mathbb{Z}^3/\ell_j\mathbb{Z}^3$ of side $\ell_j = n/L^j$ use the fluctuation space V_j and its fixed orthonormal basis, and the maps $A_j^{\min}, \mathrm{Sq}_j$ of the photon construction [26d, §3.2–3.3]. The map $A_j^{\min}$ is the constrained photon minimizer, and Sq_j is the fluctuation map formed from the positive covariance square root and the coordinate embedding into constrained bond fields. We write μ_I for the standard Gaussian on V_j and $\mathrm{Box}_j = \{\zeta_j : |\zeta_j(\Upsilon)| \le p_{0,j}$ for every basis coordinate $\Upsilon\}$. For a real terminal unit bond field A and $\zeta = (\zeta_j)_{j<k}$ define

$$A_k^{\downarrow} = A, \qquad A_j^{\downarrow} = A_j^{\min}(A_{j+1}^{\downarrow}) + \mathrm{Sq}_j\zeta_j, \qquad j = k-1, \dots, 0. \tag{8}$$

The last field is denoted $A_0^{\downarrow}(A, \zeta)$. These fields are linear in their data.

For any fine unit field A_0 define its block fields by $A_j = (\mathbf{Q}^{(j)}A_0)^{[j]}$, where $\mathbf{Q}^{(j)}$ is the j-fold photon average on the original unit torus and $B^{[j]}(b) = L^{j/2}B(L^j b)$. The block-averaging/scaling identities give $(A_{j+1})_L = \mathbf{Q}A_j$. Let Ax_k be the fields A_0 for which $\tau A_j = 0$ for $j < k$. For each unit torus put $V_{\mathrm{t}} = \delta C^2$, the real coclosed bond fields with mean zero. On the fine torus let $F_N = dC^1(n)$ with its counting inner product and standard Gaussian γ_N. The slice is

$$\mathrm{Sl}_k = \{A_0 \in \mathrm{Ax}_k : A_k \in V_{\mathrm{t}}(T_k^0)\}.$$

We will use the elementary cochain identities

$$C^1 = dC^0 \oplus \{\text{constant bond fields}\} \oplus \delta C^2, \qquad \ker d = dC^0 \oplus \{\text{constant bond fields}\}. \tag{9}$$

The sum is orthogonal. Indeed $dd = 0$, so dC^0 and δC^2 are orthogonal, and their common orthogonal complement is $\ker(d\delta + \delta d)$. Commuting lattice differences make this operator the scalar Laplacian in each component. Its kernel consists of constants: the energy identity forces every nearest-neighbor difference to vanish. This proves (9). Consequently d is injective on V_{t} and, on a unit torus with ℓ^3 sites, $\dim V_{\mathrm{t}} = \dim dC^1 = 2\ell^3 - 2$.

Proposition 3.1. The map $(A, \zeta) \mapsto A_0^{\downarrow}(A, \zeta)$ is a linear bijection from $C^1(T_k^0) \times \prod_{j<k} V_j$ to Ax_k. The block fields of its image are exactly $A_j^{\downarrow}$. The coboundary is a linear bijection $d : \mathrm{Sl}_k \to F_N$. Thus curvature has coordinates

$$\mathcal{J}_{\mathrm{ph}} : F \longmapsto (A_*; \zeta_0, \dots, \zeta_{k-1}), \qquad F = dA_0^{\downarrow}(A_*, \zeta), \tag{10}$$

where $A_* \in V_{\mathrm{t}}(T_k^0)$. They satisfy the exact energy identity

$$\|F\|^2 = \|dH_k^{\mathrm{ph}}A_*\|_2^2 + \sum_{j<k} \|\zeta_j\|^2. \tag{11}$$

The image of γ_N under $\mathcal{J}_{\mathrm{ph}}$ is

$$\mathsf{dm}_*(A_*) \prod_{j<k} \mathrm{d}\mu_I(\zeta_j), \tag{12}$$

where the terminal centered Gaussian on $V_{\mathrm{t}}(T_k^0)$ is nondegenerate and has density proportional to

$$\exp(-\|dH_k^{\mathrm{ph}}A\|_2^2/2).$$

There are orthonormal vectors $\phi_{j,\Upsilon} \in F_N$ such that $\zeta_j(\Upsilon) = \langle \phi_{j,\Upsilon}, F\rangle$. For each terminal unit plaquette p there is $\phi_p \in F_N$ with

$$(dA_*)(p) = \langle \phi_p, F\rangle, \qquad \|\phi_p\| \le 1. \tag{13}$$

There are $3\ell_*^3$ such plaquettes, and

$$\dim V_j = (2 - 2L^{-3})\ell_j^3 \le 2\ell_*^3 L^{3(k-j)}. \tag{14}$$

■

Proof. The one-step minimizer and fluctuation maps obey $\mathbf{Q}A_j^{\min}(A') = A'_L$, $\tau A_j^{\min}(A') = 0$, $\mathbf{Q}\mathrm{Sq}_j = 0$ and $\tau\mathrm{Sq}_j = 0$. Apply $\mathbf{Q}$ to (8); the injective scaling map then identifies the next block field with $A_{j+1}^{\downarrow}$. Starting from $A_0 = A_0^{\downarrow}$ proves this at all levels, including $\tau A_j = 0$. Conversely, for $A_0 \in \mathrm{Ax}_k$ its difference $A_j - A_j^{\min}(A_{j+1})$ is annihilated by $\mathbf{Q}$ and τ. The coordinate bijection [26d, (21)] and the invertible square root give exactly one ζ_j with this difference equal to $\mathrm{Sq}_j\zeta_j$. Downward recursion recovers A_0. This proves the first bijection and its inverse.

The orthogonality [26d, (28)] and the Gaussian fluctuation decomposition give at each level

$$\|dH_j^{\mathrm{ph}}A_j\|_2^2 = \|dH_{j+1}^{\mathrm{ph}}A_{j+1}\|_2^2 + \|\zeta_j\|^2.$$

Summing, and using $H_0^{\mathrm{ph}} = I$, proves (11). The dimension of the slice is

$$2\ell_k^3 - 2 + \sum_{j<k}(2\ell_j^3 - 2\ell_{j+1}^3) = 2n^3 - 2 = \dim F_N.$$

If $dA_0 = 0$ on the slice, the energy identity forces every $\zeta_j = 0$ and $dH_k^{\mathrm{ph}}A_k = 0$. Since $\mathbf{Q}_k H_k^{\mathrm{ph}} = I$ and $d\mathbf{Q}_k = \mathbf{Q}_k^{(2)}d$, this implies $dA_k = 0$. The injectivity of d on V_{t} gives $A_k = 0$, whence $A_0 = 0$ by the downward bijection. Equality of dimensions therefore proves that $d : \mathrm{Sl}_k \to F_N$ is bijective and gives (10).

Equip the coordinate space with the scalar product

$$\langle (A,\zeta),(A',\zeta')\rangle_{\mathrm{ph}} = \langle A, \Delta_k A'\rangle + \sum_{j<k}\langle \zeta_j, \zeta'_j\rangle.$$

Its terminal part is positive definite: $\langle A, \Delta_k A\rangle = \|dH_k^{\mathrm{ph}}A\|_2^2 \geq \|dA\|_2^2 > 0$ for nonzero $A \in V_{\mathrm{t}}$. Here the block curvature contraction follows from [26d, (5)]. The energy identity and polarization make $\mathcal{J}_{\mathrm{ph}}$ a linear isometry for this scalar product. It maps orthonormal bases to orthonormal bases; the Jacobian is therefore 1 for their Euclidean volume measures. These are the measures in the normalized standard Gaussians. In an orthogonal direct sum, that volume measure and the Gaussian density factor into those of the summands. This proves (12), including its normalization. The volume measure for the terminal scalar product differs from the ordinary Lebesgue measure on V_{t} by one constant factor. Thus its Gaussian has exactly the stated density. Its quadratic form is positive definite, so it is nondegenerate.

For a basis vector Υ in V_j, let $\mathbf{e}_{j,\Upsilon}$ be the coordinate vector with that single nonzero entry. Its inverse image under the isometry is $\phi_{j,\Upsilon}$. The coordinate vectors are orthonormal, proving orthonormality and the required coordinate formula. The terminal curvature functional satisfies

$$|(dA_*)(p)| \leq \|dA_*\|_2 = \|\mathbf{Q}_k^{(2)}dH_k^{\mathrm{ph}}A_*\|_2 \leq \|dH_k^{\mathrm{ph}}A_*\|_2 \leq \|F\|.$$

It is consequently represented by a vector of norm at most 1 in the finite Euclidean space F_N, giving (13). Finally [26d, (19)] and $\ell_j = L^{k-j}\ell_*$ give (14). □

If a terminal field A^c is closed, linearity gives the difference of downward fields as $A_0^{\downarrow}(A^c, 0)$. For a closed input the background identity [26d, (30)] gives zero curvature energy at the preceding level. Applying the average shows that its block minimizer is closed. The identity $\mathbf{Q}A_j^{\min}(A') = A'_L$ and preservation of means by the block average give $\mathrm{mean}(A_j^{\min}(A')) = L^{-1/2}\,\mathrm{mean}(A')$. Its final mean is $L^{-k/2}\,\mathrm{mean}(A^c)$. Equation (9) therefore gives

$$A_0^{\downarrow}(A + A^c, \zeta) = A_0^{\downarrow}(A,\zeta) + L^{-k/2}\,\mathrm{mean}(A^c) + d\lambda. \tag{15}$$

In particular adding a coarse gradient adds a fine gradient. For the harmonic twist $c_{\theta,\mu} = \theta_\mu/(e_*\ell_*)$, $L^{-k/2}c_\theta = H_\theta$, where $H_{\theta,\mu} = \theta_\mu/(qn)$. Thus its downward shift is $H_\theta + d\lambda$.

3.2. **The terminal density with an external connection.** Let $(m_{\mathrm{in}}, \varepsilon_{\mathrm{in}}^{\mathrm{v}})$ be a controlled real pair on R, with the trajectory of [26a, §7.1]. Thus its initial activity is zero, the actual block recursion exists through $k = k_*(R)$, and $|m_j| \le w_j^{\mathrm{m}} = e_j^{3/4-8\varepsilon}$ for $j \le k$. Its activity satisfies $\|E_j\|_j \le \|\mathscr{E}_j\|_j^x \le 4e_j^{1/4-6\varepsilon} \le w_j^{\mathrm{E}} = e_j^{1/4-7\varepsilon}$ by [26a, Proposition 7.1]. We retain the block field domains and the weighted polymer norms of [26g, §2.2–2.3] and [26i, §2.3, §5.1]. The fine mesh at level j is $\eta_j = L^{-j}$. In particular,

$$\mathrm{Dom}_j = \{A \text{ real unit bond field} : H_j^{\mathrm{ph}} A \in \tfrac{1}{4}\widetilde{R}_j\},$$
$$\mathfrak{X}_j = \{\mathscr{C} : \|\mathscr{C}\|_\infty < e_j^{-1+2\varepsilon},\ \|\partial\mathscr{C}\|_\infty < e_j^{-1+3\varepsilon}\}.$$

The domain $\widetilde{R}_j$ allows a global complex small field and local real-gauge representatives with the three bounds $e_j^{-3/4+i\varepsilon}$, $i = 1, 2, 3$, in the supremum, first-difference and Hölder first-difference norms; its precise enlarged cubes are those of [26d, §6]. An external connection is a complex fine unit bond field a satisfying $a^{[j]} \in \mathfrak{X}_j$ for every $j \le k$; here $a^{[j]}(b) = L^{j/2} a(L^j b)$. Write $\mathscr{C}_* = a^{[k]}$. For $A \in \mathrm{Dom}_k$ put $\mathscr{A} = H_k^{\mathrm{ph}} A$, $B = \mathscr{A} + \mathscr{C}_*$, and insert the fermion fields $H_k^{\#}(B)\Psi$ and their barred partners. The notation $H_k^{\#}$ includes $H_k\Psi$ and its covariant Hölder differences, together with $\bar{H}_k\bar{\Psi}$ and their barred differences; the operators are defined in (55). The kernel $\mathscr{E}_k(X; \mathscr{A}, \mathscr{C}, \psi, \chi)$ is the resulting Grassmann activity on an M-polymer X, and $\mathfrak{P}_k(X, \mathscr{C})$ is its external scalar family. They are the actual recursively constructed families in [26i, (7)] and [26a, (55)–(57)]. The terminal function is

$$Y_*^x(A; \mathscr{C}_*) = \det D_k(0)^{-1} \exp\left(-\varepsilon_k^{\mathrm{v}} \ell_*^3 + \sum_X \mathfrak{P}_k(X, \mathscr{C}_*)\right)$$
$$\times \int \mathrm{D}\Psi \, \exp\left(-\langle\bar{\Psi}, D_k(B)\Psi\rangle - m_k\langle\bar{\psi}, \psi\rangle + \sum_X \mathscr{E}_k(X; \mathscr{A}, \mathscr{C}_*, \psi, \chi)\right). \tag{16}$$

Since $k \ge 1$, this Berezin integral has $4\ell_*^3$ pairs with weight 1. The inserted chamber fields and kernels retain their fine weight $\eta_k^3/48$. The inverse and field maps of [26b, Theorem A and §4.2], together with the activity domains and exact density of [26i, §2.3 and §3], make every term well defined. At zero connection, the positive-reference-mass inverse formula in [26i, Proposition 3.1] gives $\det D_k(0) \ne 0$. At zero external slot, $\mathfrak{P}_k(X, 0) = 0$ and $\mathscr{E}_k|_0 = E_k$; we write $Y_*(A) = Y_*^x(A; 0)$.

For the original fine unit field let

$$w_N(B) = \exp(-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3) \frac{\det(D^{\mathrm{ch}}[qB] + \bar{m}\eta + m_{\mathrm{in}})}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)}. \tag{17}$$

The matrices have size $192n^3$. The bracket argument of D^{ch} is the bond phase, as in [26e, §1.1, (6)]; an unabsorbed unit connection B therefore has argument qB. The denominator is nonzero by the free positive-mass coercivity estimate; no invertibility of the numerator is required. The weighted Gaussian Berezin identity gives the initial integrated density exactly as $w_N(A_0 + a)$. In particular all initial measure and vacuum factors are included in (17).

This fine weight is entire in B and is invariant under real gauge shifts. Its entries involve only the four chamber neighbors, with link phases in the fourth neighbor. Multiplication by $\exp(-\mathrm{i}q\lambda(c))$ on all chamber components over c conjugates the operator at B to the operator at $B + d\lambda$: the endpoint phases cancel the gradient in the fourth hopping phase, and the other hops stay over the same c. The determinants therefore agree. The same stencil bounds every row sum at $B_0 + a$, with B_0 real, by $|\boldsymbol{\mu}| + 41 + 6\exp(q\|\operatorname{Im} a\|_\infty)$, where $\boldsymbol{\mu} = \bar{m}\eta + m_{\mathrm{in}}$. Indeed the three internal hopping block norms sum with the diagonal to less than $|\boldsymbol{\mu}| + 41$, and the fourth has row norm at most $6\exp(q\|\operatorname{Im} a\|_\infty)$. The determinant expansion bounds its modulus by the product of the absolute

row sums: its permutation sum is a subsum of the sum over all column choices. Consequently

$$|w_N(B_0+a)| \le B_N(a) := \frac{\exp(-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3)}{|\det(D^{\mathrm{ch}}[0]+\bar{m}\eta)|}\big(|\boldsymbol{\mu}|+41+6\exp(q\|\operatorname{Im} a\|_\infty)\big)^{192n^3} < \infty. \tag{18}$$

This bound is used for integrability at a fixed cutoff.

Let $A(F) \in V_{\mathrm{t}}$ be the transverse fine potential of $F \in F_N$, and define the finite complex measure

$$\mathrm{d}\nu^{\theta,a}(F) = w_N(A(F)+H_\theta+a)\mathrm{d}\gamma_N(F).$$

The small-field event and the terminal indicator are

$$\mathfrak{G} = \{F : |\zeta_j(\Upsilon)| \le p_{0,j}\ (j<k),\ \|dA_*\|_\infty \le p_*\}, \qquad \chi_*(A) = \mathbf{1}[\|dA\|_\infty \le p_*]. \tag{19}$$

Here A_*, ζ are the actual coordinates (10). Products with χ_* are extended by zero off its support.

Theorem B. For every $A \in \mathrm{Dom}_k$,

$$Y_*^x(A;\mathscr{C}_*) = \int \prod_{j<k} (\mathbf{1}_{\mathrm{Box}_j}(\zeta_j)\mathrm{d}\mu_I(\zeta_j)) w_N(A_0^{\downarrow}(A,\zeta)+a). \tag{20}$$

The integrand is continuous on the compact product of boxes. If $A, A+d\lambda \in \mathrm{Dom}_k$, with λ real, their terminal functions agree. For every bounded Borel function Φ on $V_{\mathrm{t}}(T_k^0)$ and every real twist θ,

$$\nu^{\theta,a}(\mathbf{1}_{\mathfrak{G}}\Phi(A_*)) = \int \mathrm{d}\mathsf{m}_*(A)\,\Phi(A)\chi_*(A)Y_*^x(A+c_\theta;\mathscr{C}_*). \tag{21}$$

Its right-hand density is bounded and Borel, and $Y_*^x(A+c_\theta;\mathscr{C}_*)$ is jointly continuous in (θ, A) on $\{\chi_*=1\}$. ■□

Proof. The photon domain conditions [26d, (43)] hold at each running scale by the fixed charge choice. [26d, Theorem B] gives two consequences used here: a real unit field with curvature bounded by p_j lies in Dom_j; and a field in Dom_{j+1} descends to Dom_j after addition of any box fluctuation. Downward induction therefore puts every field of (8) in its respective domain whenever its top field lies in Dom_k and all fluctuations lie in their boxes.

Let $y_j^x(A;a^{[j]})$ be the integrated density at level j along this trajectory. The exact integrated step [26i, (9)] gives

$$y_{j+1}^x(A';a^{[j+1]}) = \int_{\mathrm{Box}_j} \mathrm{d}\mu_I(\zeta_j)\, y_j^x(A_j^{\min}(A')+\mathrm{Sq}_j\zeta_j;a^{[j]}).$$

Its external slot is the required one because $(a^{[j+1]})_L = a^{[j]}$, directly from the two scaling factors. Its coupling maps are the actual trajectory maps, and its vacuum includes the box normalization. The first such identity integrates the full $192n^3$ weighted chamber pairs into $4\ell_1^3$ coarse pairs.

For a top field in Dom_j, the function $w_N(A_0^{\downarrow}(A,\zeta)+a)$ is continuous in the finitely many fluctuation coordinates. Every finite product of boxes is compact, and (18) bounds it. Thus all repeated integrals are absolutely integrable. Starting with the initial identity $y_0^x(A_0;a) = w_N(A_0+a)$, use the one-step formula, the downward recursion and Fubini at each level. This proves (20) at $j=k$, with its unnormalized factors $\mathbf{1}_{\mathrm{Box}_j}\mathrm{d}\mu_I$. The continuity assertion follows from the same finite-dimensional composition. Equation (15) turns a coarse real gradient into a fine gradient; gauge invariance of w_N inside this integral then proves gauge invariance of Y_*^x.

To integrate against an arbitrary bounded Borel observable, first express the fine density in the curvature coordinates. The slice potential $A_0^{\downarrow}(A_*,\zeta)$ has mean zero: its block average has the mean of the fine field, and its top field has mean zero. Its difference from $A(F)$ is closed and has mean zero, hence is a gradient by (9). The harmonic shift identity gives

$$w_N(A(F)+H_\theta+a) = w_N(A_0^{\downarrow}(A_*+c_\theta,\zeta)+a).$$

Under the tower isometry the Gaussian becomes (12), and the event $\mathfrak{G}$ has indicator

$$\chi_*(A_*)\prod_{j<k}\mathbf{1}_{\mathrm{Box}_j}(\zeta_j).$$

The integrand is a bounded Borel function with bound $\|\Phi\|_\infty B_N(a)$. The change-of-variables identity for its image measure and Fubini therefore give

$$\nu^{\theta,a}(\mathbf{1}_{\mathfrak{G}}\Phi(A_*)) = \int \mathrm{d}\mathsf{m}_*(A)\,\Phi(A)\chi_*(A)\int\prod_{j<k}(\mathbf{1}_{\mathrm{Box}_j}\mathrm{d}\mu_I(\zeta_j))w_N(A_0^{\downarrow}(A+c_\theta,\zeta)+a).$$

Where $\chi_*(A)=1$, the field $A+c_\theta$ has the same curvature bound and belongs to Dom_k. The inner integral is $Y_*^x(A+c_\theta;\mathscr{C}_*)$ by (20), proving (21).

The inner integral defines a function on all real (θ,A), whether or not $A+c_\theta\in\mathrm{Dom}_k$. Its modulus is at most $B_N(a)$. Its integrand is continuous and uniformly continuous on a compact neighborhood of any (θ,A) times the compact box product. Integrating this uniform bound proves joint continuity. Multiplying by χ_* gives a bounded Borel function, since $\{\chi_*=1\}$ is closed. On that set it is the asserted terminal density. □

3.3. **The large-field complement.** At zero external connection write $\nu^\theta=\nu^{\theta,0}$, and let $\bar\nu$ be its average over $[-\pi,\pi]^3$ with measure $\mathrm{d}\theta/(2\pi)^3$. The real chamber operator has two adjoint spin blocks; thus its determinant at every real lattice mass is the squared modulus of the determinant of one block. Consequently ν^θ is positive. The zero-sector identity [26c, (40)] gives $0<\bar\nu(1)<\infty$ and identifies $\bar\mu=\bar\nu/\bar\nu(1)$ with the zero-flux curvature probability law.

Use the fixed constant $C_0\ge1$ of [26h, (25)], recalled in section 5. It also bounds the terminal field maps and Dirac form in section 6.2. Set

$$\varepsilon_3 = 6\ell_*^3\left(e^{-p_{0,*}/2}+e^{-p_*^2/(2C_0^2)}\right),\qquad \varepsilon_4 = 6\ell_*^3e^{-p_*^2/(2C_0^2)}. \tag{22}$$

Proposition 3.2. Suppose $v_*\ge8$, the integer $p_0\ge1$ satisfies $p_0v_*^{p_0-1}\ge28$, and $L\ge5$. Then, at every real initial pair,

$$\bar\mu(\mathfrak{G}^c)\le\varepsilon_3,\qquad \mathsf{m}_*(\chi_*=0)\le\varepsilon_4. \tag{23}$$

For fixed $(\bar m,e,R)$, these bounds are independent of the ultraviolet cutoff and of the initial pair. ■

Proof. The zero-flux domination estimate in proposition 2.1 and the rows of (13) give

$$\bar\mu\{|\zeta_j(\Upsilon)|>p_{0,j}\}\le2e^{-p_{0,j}^2/2},\qquad \bar\mu\{|(dA_*)(p)|>p_*\}\le2e^{-p_*^2/2}.$$

Using the coordinate counts in (14), the union bound is

$$\bar\mu(\mathfrak{G}^c)\le4\ell_*^3\sum_{i=1}^{k}L^{3i}e^{-(v_*+i\log L/2)^{2p_0}/2}+6\ell_*^3e^{-p_*^2/2}. \tag{24}$$

Here $i=k-j$ and $v_j=v_*+i\log L/2$. The first two terms of the binomial expansion give

$$p_{0,j}=(v_*+i\log L/2)^{p_0}\ge p_{0,*}+28i\log L/2.$$

Since $p_{0,j}\ge1$, its square is at least $p_{0,j}$. It follows that

$$L^{3i}e^{-p_{0,j}^2/2}\le e^{-p_{0,*}/2}L^{-4i}.$$

Summing the geometric series, whose sum is $(L^4-1)^{-1}\le1$, bounds (24) by $4\ell_*^3e^{-p_{0,*}/2}+6\ell_*^3e^{-p_*^2/2}$. This is at most ε_3, since $C_0\ge1$.

The image of γ_N under $F\mapsto A_*$ is m_* by (12). For each plaquette the variable $\langle\phi_p,F\rangle$ is Gaussian with variance $\|\phi_p\|^2\le1$. Its two-sided exponential Markov bound is $2e^{-p_*^2/2}$. Summing over the $3\ell_*^3$ plaquettes proves $\mathsf{m}_*(\chi_*=0)\le6\ell_*^3e^{-p_*^2/2}\le\varepsilon_4$. Finally, ℓ_* and e_* are independent of N after the mesh threshold, by [26a, Proposition 6.1]; hence the displayed errors have the asserted dependence. □

4. Free pressure and harmonic twists

The partition comparison requires an error in the free pressure which depends on the physical size, rather than on the number of lattice sites. It also requires a twist estimate for the operator with a block penalty. The latter penalty removes the small-mass singularity before the fine mesh is taken to zero.

For $0 < m \le 1$ let $\mathfrak{d}_m(p) = \det \mathcal{D}_m(p)$, where $\mathcal{D}_m$ is the 192-dimensional chamber symbol in [26b, §2.2], and set

$$p^0_{\mathrm{ch}}(m) = \int_{[-\pi,\pi]^3} \log \mathfrak{d}_m(p) \frac{\mathrm{d}p}{(2\pi)^3}. \tag{25}$$

The spin-block identity and positive real part imply $\mathfrak{d}_m(p) > 0$ for real p. In this subsection $|z| = |z|_\infty$ for $z \in \mathbb{Z}^3$. Write $\kappa_{(0)} = \kappa_{\mathrm{strip}}$ from [26b, Proposition 2.2].

Proposition 4.1. There is a finite C_6, depending only on ρ, such that, with

$$\varepsilon_5(x) = C_6 \log \frac{1}{1 - e^{-\kappa_{(0)} x}} \qquad (x > 0), \tag{26}$$

the free twisted determinant on a unit torus of side $n \ge 1$ satisfies

$$\left| \log \det(D^{\mathrm{ch}}[qH_\theta] + m) - n^3 p^0_{\mathrm{ch}}(m) \right| \le \varepsilon_5(mn). \tag{27}$$

Here $H_{\theta,\mu} = \theta_\mu/(qn)$ and $\theta \in \mathbb{R}^3$. For $m = \bar{m}\eta$ and $n = R/\eta$ define

$$z^{\mathrm{free}}_{r,n} = \int_{[-\pi,\pi]^3} \det(D^{\mathrm{ch}}[qH_\theta] + \bar{m}\eta) \frac{\mathrm{d}\theta}{(2\pi)^3}.$$

Then $z^{\mathrm{free}}_{r,n} > 0$ and

$$\left| \log z^{\mathrm{free}}_{r,n} - n^3 p^0_{\mathrm{ch}}(\bar{m}\eta) \right| \le \varepsilon_5(\bar{m}R). \tag{28}$$

■

Proof. The plane waves with momenta $P_{n,\theta} = \{(2\pi j + \pi \mathbf{1} + \theta)/n : j \in \{0, \dots, n-1\}^3\}$ diagonalize the translation part of the chamber operator. Inserting a plane wave in the fourth hop shifts its momentum by θ/n, whereas the internal hops are unchanged. Thus

$$\det(D^{\mathrm{ch}}[qH_\theta] + m) = \prod_{p \in P_{n,\theta}} \mathfrak{d}_m(p) > 0. \tag{29}$$

The phases contain $qH_\theta = \theta/n$, so the determinant is independent of q.

Factor the determinant by the light–heavy decomposition:

$$\log \mathfrak{d}_m(p) = f_H(p) + f_S(p), \qquad f_H(p) = \log\det D_{\mathrm{hh}}(p), \quad f_S(p) = \log\det B_m(p).$$

Both logarithms are real and smooth on real momenta. Their derivatives, which suffice for contour shifting, are

$$\partial_\mu f_H = \mathrm{tr}(D^{-1}_{\mathrm{hh}} \partial_\mu D_{\mathrm{hh}}), \qquad \partial_\mu f_S = \mathrm{tr}(B_m^{-1} \partial_\mu B_m).$$

The heavy block has dimension 188. On the strip of width θ_0 its inverse has norm at most $2/g_{\mathrm{h}}$, and the fourth-hop derivative has norm at most $3(1+\rho)e^{\theta_0} = 3(1+\rho) + g_{\mathrm{h}}/2$. Hence

$$|\partial_\mu f_H| \le C_{\mathrm{lH}} := 188 \left(1 + \frac{6(1+\rho)}{g_{\mathrm{h}}}\right).$$

For $z \ne 0$ choose $|z_\mu| = |z|$. One integration by parts and a shift of that contour by $-\mathrm{i}\theta_0 \operatorname{sgn}(z_\mu)$ give

$$|\widehat{f}_H(z)| \le C_{\mathrm{lH}} |z|^{-1} e^{-\theta_0 |z|}. \tag{30}$$

The derivative trace is holomorphic in a neighborhood of the closed strip. Its periodicity cancels the vertical sides of the contour rectangle; no choice of a complex logarithm is needed.

For the light factor put $r_m(p') = (|p'|_2^2+m^2)^{1/2}$. The inverse derivative bounds [26b, (15)] hold on $|\operatorname{Im} p_\nu| \le \kappa_{(0)}m$. The bound K_s for B_m on the strip of width θ_0 and Cauchy's formula on polydiscs of radius $\theta_0/2$ give $\|\partial^\beta B_m\| \le 24K_s(2/\theta_0)^{|\beta|}$ for $1 \le |\beta| \le 4$. Leibniz's rule in the derivative trace therefore gives

$$|\partial^\alpha \partial_\mu f_S(p'+\mathrm{i}p'')| \le C_{\mathrm{lS}}\, r_m(p')^{-1-|\alpha|} \le C_{\mathrm{lS}}|p'|_2^{-1-|\alpha|} \quad (|\alpha| \le 3,\ p' \ne 0), \tag{31}$$

where one may take

$$C_{\mathrm{lS}} = \max_{|\alpha|\le 3} 96K_s \sum_{\beta\le\alpha} \binom{\alpha}{\beta} C_\beta(\rho)(2/\theta_0)^{|\alpha|-|\beta|+1}(3\pi^2+1)^{(|\alpha|-|\beta|)/2}.$$

This constant is independent of m. Shift the differentiated Fourier integral by $-\mathrm{i}\kappa_{(0)}m\,\mathrm{sgn}(z_\mu)e_\mu$. The periodic Fourier estimate proved in [26b, Proposition 2.3], with singular order 1 and three derivatives, bounds this integral by $c_{\mathrm{F}}C_{\mathrm{lS}}(1+|z|)^{-2}$, where c_{F} is its constant at those orders. Thus

$$|\widehat{f}_S(z)| \le c_{\mathrm{F}}C_{\mathrm{lS}}|z|^{-1}(1+|z|)^{-2}e^{-\kappa_{(0)}m|z|}. \tag{32}$$

The strip is contained in the heavy strip since $\kappa_{(0)}m \le \theta_0/2$. The inverse is holomorphic near the closed strip, so the contour shift is legitimate.

The Fourier coefficients of $f = f_H + f_S$ are summable. For a summable periodic Fourier series, summing over $P_{n,\theta}$ keeps precisely the indices $z = nw$:

$$\sum_{p\in P_{n,\theta}} f(p) = n^3 \sum_{w\in\mathbb{Z}^3} \widehat{f}(nw)(-1)^{w_1+w_2+w_3}e^{\mathrm{i}\theta\cdot w}. \tag{33}$$

This follows by summing the geometric progression in each coordinate; absolute convergence permits the interchange. Its $w=0$ term is $n^3p^0_{\mathrm{ch}}(m)$. There are $24t^2+2 \le 26t^2$ indices with $|w|=t$. The contribution of (32) is at most

$$26c_{\mathrm{F}}C_{\mathrm{lS}} \sum_{t\ge1} t^{-1}e^{-\kappa_{(0)}mnt}.$$

For the heavy term use $\kappa_{(0)}m \le \theta_0/2$ and $x^2e^{-\theta_0 x/2} \le 16/(\exp(1)\theta_0)^2$. Its contribution is at most

$$\frac{416C_{\mathrm{lH}}}{(\exp(1)\theta_0)^2} \sum_{t\ge1} t^{-1}e^{-\kappa_{(0)}mnt}.$$

Since $\sum_{t\ge1} e^{-xt}/t = \log(1-e^{-x})^{-1}$, (27) follows with $C_6 = 26[c_{\mathrm{F}}C_{\mathrm{lS}} + 16C_{\mathrm{lH}}/(\exp(1)\theta_0)^2]$. Finally every twisted determinant lies between $\exp(n^3p^0_{\mathrm{ch}}(m) \pm \varepsilon_5(mn))$. Its harmonic average lies in the same positive interval. Taking logarithms proves (28). □

We next retain the block penalty. Let $\mathfrak{N} \ge 3$ be odd and put $\tau = \mathfrak{N}^{-1}$. On the fine torus of side $\ell \in \mathbb{N}$ let $\mathcal{H}_\theta$ be the chamber spinors with mesh τ and boundary multipliers $-e^{\mathrm{i}\theta_\mu}$, where $\theta \in \mathbb{R}^3$. The scalar product uses $\tau^3/48$. The coarse space has four components per unit site and counting scalar product. The average Q takes the mean over each unit block and its 48 labels, and Q^* extends a coarse field constantly on that block. Thus $QQ^* = 1$, $\|Q\| = \|Q^*\| = 1$, and $\Pi_{\mathrm{bl}} = Q^*Q$ is an orthogonal projection. Fix $0 < b_- \le b_+ < \infty$; the block construction uses $b_- = 1 - L^{-1}$ and $b_+ = 1$. Let D be the free chamber Dirac operator at mesh τ on $\mathcal{H}_\theta$. For $0 < \mu \le 1$ and $b_- \le b \le b_+$ set

$$T_\theta = D + \mu + b\Pi_{\mathrm{bl}}, \qquad D^{\mathrm{bl}}_\theta = b - b^2QT_\theta^{-1}Q^*. \tag{34}$$

All determinants below are determinants of linear maps. The weighting of a scalar product changes their matrix representations, but does not add a determinant factor.

Proposition 4.2. The determinants of T_θ, $D+\mu$, and D^{bl}_θ are positive, and

$$(D^{\mathrm{bl}}_\theta)^{-1} = b^{-1} + Q(D+\mu)^{-1}Q^*, \tag{35}$$

$$\det D_\theta^{\mathrm{bl}} = b^{4\ell^3} \frac{\det(D+\mu)}{\det T_\theta}. \tag{36}$$

There are $C_\Sigma < \infty$ and $0 < \gamma_\Sigma \le 1$, depending only on (b_-, b_+, ρ), such that

$$\left| \log \frac{\det T_\theta}{\det T_0} \right| \le C_\Sigma \ell^{-1} e^{-\gamma_\Sigma \ell}. \tag{37}$$

For the level $k \ge 1$ data

$$(\mathfrak{N}, \ell, \mu, b) = (L^k, \ell_*, \mu_*, b_k), \qquad c_{\theta,\mu} = \theta_\mu/(e_*\ell_*),$$

put $R_\theta^{\mathrm{tw}} = \det D_k(c_\theta)/\det D_k(0)$. Then

$$R_\theta^{\mathrm{tw}} = \frac{\det(D^{\mathrm{ch}}[qH_\theta] + \bar{m}\eta)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)} \frac{\det T_0}{\det T_\theta} > 0. \tag{38}$$

■

Proof. The real part of $D+\mu$ is at least μ. The same holds for T_θ, since its penalty is positive. Both operators have two adjoint spin blocks; their determinants are therefore positive. Put $G = (D+\mu)^{-1}$, $S = QGQ^*$ and $Z = (b^{-1} + S)^{-1}$. Their inverses exist since $\operatorname{Re} G \ge 0$ and $\operatorname{Re}(b^{-1} + S) \ge b^{-1}$. Multiplication gives $T_\theta^{-1} = G - GQ^*ZQG$ and $D_\theta^{\mathrm{bl}} = Z$, proving (35). Sylvester's determinant identity, applied to $-bQ$ and $T_\theta^{-1}Q^*$, gives (36). Its right side is positive.

We prove (37) with constants independent of τ and μ. Let

$$c_{\mathrm{nz}} = \min\left\{ \frac{\rho}{\pi}, \frac{g_{\mathrm{h}}\rho^2}{49\sqrt{3}\,\pi^3} \right\} > 0.$$

For real $q' \in [-\pi, \pi]^3$,

$$\|\mathcal{D}_0(q')u\| \ge c_{\mathrm{nz}} |q'|_2 \|u\|. \tag{39}$$

Normalize $\|u\| = 1$ and write $u = Pu + \Pi_{\mathrm{h}} u$ for its light and heavy parts, where $\Pi_{\mathrm{h}} = 1 - P$ is the cluster-heavy projection. If $d = \|\mathcal{D}_0(q')u\|$, then $g_{\mathrm{h}}\|\Pi_{\mathrm{h}}u\|^2 \le d$. The light compression is $W(q') = \rho\sum_\mu (1 - \cos q'_\mu) + \mathrm{i}\gamma \cdot \sin q'$; its squared modulus is at least $(2\rho/\pi)^2 |q'|_2^2$. Also $J^*\mathcal{D}_0(0) = 0$, and the fourth-hop difference has norm at most $6|q'|_2$. Projecting $\mathcal{D}_0(q')u$ onto the light space gives

$$d \ge |q'|_2 \left(\frac{2\rho}{\pi} - 7\|\Pi_{\mathrm{h}} u\| \right).$$

If $\|\Pi_{\mathrm{h}}u\| \le \rho/(7\pi)$ this is at least $(\rho/\pi)|q'|_2$. Otherwise $d \ge g_{\mathrm{h}}\rho^2/(49\pi^2)$; since $|q'|_2 \le \sqrt{3}\pi$, this proves (39).

On one unit cell use the Bloch space $\mathcal{H}(p) = \{f(c + e_\mu, g) = e^{\mathrm{i}p_\mu} f(c, g)\}$. Its modes are $p_j = p + 2\pi j$, with $j \in I_{\mathfrak{N}}^3$, $I_{\mathfrak{N}} = \{-(\mathfrak{N}-1)/2, \dots, (\mathfrak{N}-1)/2\}$. The map $\{u_j\} \mapsto \sum_j e^{\mathrm{i}p_j \cdot c} u_j(g)$ is unitary for the one-cell scalar product $\tau^3/48$. It diagonalizes D with blocks $X_j = \tau^{-1}\mathcal{D}_0(\tau p_j)$. Define

$$u_1(t) = \mathfrak{N}^{-1} \sum_{s \in I_{\mathfrak{N}}} e^{-\mathrm{i}ts/\mathfrak{N}}, \qquad u(p) = \prod_\mu u_1(p_\mu).$$

The block projection in this representation has kernel $u(p_j)u(p_{j'})P$. This follows by averaging the plane waves on the centered unit block; discrete Fourier orthogonality expands the constant block extension. For $p \in [-\pi, \pi]^3$ one has $\tau p_j \in [-\pi, \pi]^3$ and $|p_j|_2 \ge \pi|j|$ when $j \ne 0$. Hence (39) gives $\|X_j u\| \ge c_{\mathrm{nz}} |p_j|_2 \|u\|$.

The inverse on $\mathcal{H}(p)$ is uniformly bounded. To see this, let $\|f\| = 1$ and $d = \|(D+\mu+b\Pi_{\mathrm{bl}})f\| \le 1$. The real-part estimate gives

$$\mu \le d, \qquad \|\Pi_{\mathrm{bl}} f\| \le (d/b_-)^{1/2}, \qquad \|\Pi_{\mathrm{h}} f\| \le (d/(3g_{\mathrm{h}}))^{1/2}, \qquad \|Df\| \le (2 + \sqrt{b_+})\sqrt{d}.$$

Separate the mode $j = 0$ from f^{r}, the remaining modes. The bound on X_j gives $\|f^{\mathrm{r}}\| \le \|Df\|/(\pi c_{\mathrm{nz}})$. Moreover $u_1(t) = \sin(t/2)/[\mathfrak{N}\sin(t/(2\mathfrak{N}))] \ge 2/\pi$ for $|t| \le \pi$, with value 1 at zero.

Thus the block projection of the zero mode controls its light part by the factor $8/\pi^3$. Using $\|\Pi_{\mathrm{bl}}\| = 1$ and subtracting the other modes,

$$\sqrt{d/b_-} \geq \frac{8}{\pi^3}(1 - \|\Pi_{\mathrm{h}} f\|) - (1 + 8/\pi^3)\|f^{\mathrm{r}}\|.$$

It follows that $8/\pi^3 \leq K_{\mathrm{iv}}\sqrt{d}$, where

$$K_{\mathrm{iv}} = b_-^{-1/2} + \frac{8}{\pi^3}(3g_{\mathrm{h}})^{-1/2} + (1 + 8/\pi^3)\frac{2 + \sqrt{b_+}}{\pi c_{\mathrm{nz}}}.$$

Consequently the inverse norm is at most $C_{\mathrm{iv}} = \max\{1, (\pi^3 K_{\mathrm{iv}}/8)^2\}$.

For complex momenta use the periodic gauge $f(c, g) \mapsto e^{\mathrm{i}p\cdot c} f(c, g)$ to work on the fixed periodic one-cell space. Let $D(p)$ and $\Pi(p)$ be the resulting operators, so $\mathcal{T}(p) = D(p) + \mu + b\Pi(p)$. The fourth hopping phase is $e^{\mathrm{i}\tau p\cdot g e_1}$, and the block kernel is $e^{-\mathrm{i}p\cdot(c-c')}$. A shift of p_ν by 2π conjugates these operators by multiplication by $e^{2\pi \mathrm{i} c_\nu}$. Thus their determinant and derivative traces are periodic. For $p = p' + \mathrm{i}te_\nu$, $|t| \leq 1$, and $1 \leq j \leq 4$, direct differentiation gives

$$\|\partial_\nu^j D(p)\| \leq 9\tau^{j-1}, \qquad \|\partial_\nu^j \Pi(p)\| \leq \exp(1), \qquad \operatorname{rank} \partial_\nu^j \Pi(p) \leq 4(j+1). \tag{40}$$

The last bound follows by expanding $(c_\nu - c'_\nu)^j$ into $j+1$ products of a function of c and a function of c', each with four spin components. The norm estimates use $|c_\nu - c'_\nu| < 1$ in one block, $\tau \leq 1/3$, and $\|M_3\| \leq 6$. The difference from the real momentum has norm at most $(6 + b_+)|t|e^{|t|}$. With

$$\gamma_\Sigma = \min\{1, [2\exp(1)(6 + b_+)C_{\mathrm{iv}}]^{-1}\},$$

a Neumann series gives $\|\mathcal{T}(p)^{-1}\| \leq 2C_{\mathrm{iv}}$ on this strip. Its singular values are at most twice those at p'.

The growing dimension of the fine cell cannot be used in a trace bound. Instead write $G(p) = \mathcal{T}(p)^{-1}$ and compare at real p' with $(D(p') + 1)^{-1}$. The resolvent identity, the bound just proved, and the singular-value multiplication inequality [Pon25, slides 3–9] imply

$$s_j(G(p)) \leq C_{\mathrm{sv}} s_j((D(p') + 1)^{-1}), \qquad C_{\mathrm{sv}} = 2[1 + (1 + b_+)C_{\mathrm{iv}}].$$

The comparison operator is the direct sum of $(X_j + 1)^{-1}$. Since $\operatorname{Re} X_j \geq 0$, each of its 192 singular values is at most $(1 + c_{\mathrm{nz}}^2 |p'_j|_2^2)^{-1/2}$. Set

$$S_4 = \sum_{j \in \mathbb{Z}^3 \setminus \{0\}} |j|^{-4} = 4\pi^2 + \pi^4/45, \quad C_{\mathrm{uv}} = (1 + 3\pi^2 c_{\mathrm{nz}}^2)^{1/2}, \quad S_{\mathrm{jf}} = 1 + S_4/(\pi c_{\mathrm{nz}})^4.$$

For $1 \leq s \leq 4$ we obtain

$$\tau^{4-s}\|G(p)\|_{\mathfrak{S}_s}^s \leq 192 C_{\mathrm{sv}}^s C_{\mathrm{uv}}^{4-s} S_{\mathrm{jf}}. \tag{41}$$

Indeed $\tau^2(1 + c_{\mathrm{nz}}^2|p'_j|_2^2) \leq C_{\mathrm{uv}}^2$, so multiplication by τ^{4-s} makes each summand a multiple of $(1 + c_{\mathrm{nz}}^2|p'_j|_2^2)^{-2}$. The nonzero indices are bounded by $(\pi c_{\mathrm{nz}}|j|)^{-4}$ and have the summable majorant displayed above.

Let $F(p) = \log\det \mathcal{T}(p)$ on real momenta. The fourth derivative is the trace expression

$$\begin{aligned} F_4(p) = {} & \operatorname{tr}(GT_4) - 4\operatorname{tr}(GT_1GT_3) - 3\operatorname{tr}(GT_2GT_2) \\ & + 12\operatorname{tr}(GT_1GT_1GT_2) - 6\operatorname{tr}((GT_1)^4), \end{aligned} \tag{42}$$

where $T_j = \partial_\nu^j \mathcal{T}$. This follows by differentiating $\partial_\nu F = \operatorname{tr}(GT_1)$ and $\partial_\nu G = -GT_1G$, using trace cyclicity. The expression is holomorphic near the closed strip in the ν coordinate. Each trace in (42) has $s \leq 4$ inverse factors and derivatives of total order four. Expand every T_j as $D_j + b\Pi_j$. For the term with only D_j, Schatten Hölder and (40) give the bound $9^s\tau^{4-s}\|G\|_{\mathfrak{S}_s}^s$. Equation (41) bounds this by $9^4\, 192 C_{\mathrm{sv}}^4 C_{\mathrm{uv}}^3 S_{\mathrm{jf}}$. In each other term estimate one $b\Pi_j$ in trace norm: its rank bound gives $\|b\Pi_j\|_{\mathfrak{S}_1} \leq 20\exp(1)b_+$. The other factors have operator norms at most $2C_{\mathrm{iv}}$ and $\max\{9, \exp(1)b_+\}$.

There are at most 15 such terms in any trace. Since the five numerical coefficients have absolute sum 26, this proves $|F_4(p)| \le C_{\mathrm{Flg}}$, where

$$C_{\mathrm{Flg}} = 26\left[9^4\,192 C_{\mathrm{sv}}^4 C_{\mathrm{uv}}^3 S_{\mathrm{jf}} + 300\exp(1) b_+ (2C_{\mathrm{iv}})^4 \max\{9, \exp(1) b_+\}^3\right].$$

All Schatten applications are on a finite-dimensional space, with s factors of exponent s and reciprocal exponents summing to one [Pon25, slides 54–56].

For $z \neq 0$ choose $|z_\nu| = |z|$. Four integrations by parts and a contour shift by $-\mathrm{i}\gamma_\Sigma \operatorname{sgn}(z_\nu) e_\nu$ now give $|\widehat{F}(z)| \le C_{\mathrm{Flg}} |z|^{-4} e^{-\gamma_\Sigma |z|}$. The trace expression is periodic by the preceding conjugacy and is holomorphic near the closed strip. Decomposition into unit-cell Bloch spaces gives $\log\det T_\theta = \sum_{p \in P_{\ell,\theta}} F(p)$. Using (33) at θ and at zero, and subtracting, proves

$$\left|\log \frac{\det T_\theta}{\det T_0}\right| \le 2\ell^3 \sum_{w \neq 0} C_{\mathrm{Flg}} (\ell|w|)^{-4} e^{-\gamma_\Sigma \ell |w|} \le 2 S_4 C_{\mathrm{Flg}} \ell^{-1} e^{-\gamma_\Sigma \ell}.$$

Thus $C_\Sigma = 2 S_4 C_{\mathrm{Flg}}$.

Finally constant connections are twists by multiplication by $e^{\mathrm{i}\theta\cdot c/\ell_*}$. Every rectilinear path contributing to the average has the same constant-connection integral, so its phase is $e^{\mathrm{i}\theta\cdot(c-y)/\ell_*}$. Consequently the covariant chamber averages and fine inverse are conjugate to Q, Q^* and T_θ^{-1}, and $D_k(c_\theta)$ is conjugate to D_θ^{bl}. Dilating the mesh τ to one gives

$$\det(D + \mu \text{ on } \mathcal{H}_\theta) = \tau^{-192 n^3} \det(D^{\mathrm{ch}}[q H_\theta] + \tau\mu), \qquad n = \ell_* \mathfrak{N}.$$

Here $\tau\mu_* = \bar{m}\eta$. Divide (36) at the two twists. The factors $b^{4\ell_*^3}$ and τ^{-192n^3} cancel, proving (38). □

5. The terminal remainder and partition comparison

Use the constant already fixed in the construction [26h, (25)]:

$$C_0 = \max\{C_P^{\mathrm{ph}} c_{\mathrm{bd}}(\gamma_P),\, C^F,\, 4b_+ + 16 C^F c(\gamma^F)\}, \qquad b_+ = 1,$$

where $c(\gamma) = \sum_{z\in\mathbb{Z}^3} e^{-\gamma|z|_\infty}$ and $c_{\mathrm{bd}}(\gamma) = 3e^{3\gamma} c(\gamma)$. Here $C_P^{\mathrm{ph}}, \gamma_P$ are the photon kernel constants and C^F, γ^F are the fermion constants. Thus $C_0 \ge 1$ bounds the photon row sums, both field maps, and the Dirac form proved in section 6.2. Put $b_- = 1 - L^{-1}$ and

$$c_\kappa = \sup_{\square} \sum_{X \supset \square} e^{-\kappa d_M(X)} \le K_0.$$

The fixed scheme gives the following inequalities at every allowed stop:

$$\begin{aligned} 8C_0^2 \ell_*^6 (b_-^{-1} + \mu_*^{-1}) &\le e_*^{-\varepsilon/2}, \\ 4(2C_0 + c_\kappa + 5c_{\mathrm{fl}} + 6)\ell_*^3 &\le e_*^{-\varepsilon}, \\ e_*^{1/4 - 9\varepsilon} \le \tfrac{1}{8}, \qquad 4C_0 \ell_*^2 p_* &< e_*^{-\varepsilon}, \\ 12\ell_*^3 \left(e^{-p_{0,*}/2} + e^{-p_*^2/(2C_0^2)}\right) \le e_*, \qquad C_\Sigma e^{-\gamma_\Sigma \ell_*} &\le e_*. \end{aligned} \tag{43}$$

Indeed, the scalar inventory [26h, (29)] holds for every $v \ge -\log e_1$. Write $\ell(v) = LMv^2$. By (2), $v_* \ge -\log e_1$, $\ell_* < \ell(v_*)$, $\mu_*^{-1} < L\ell(v_*)$, and $\ell_* \ge M v_*^2$. At $v = v_*$ the covariance condition follows from

$$8C_0^2 \ell_*^6 (b_-^{-1} + \mu_*^{-1}) \le 8C_0^2 \ell(v_*)^6 (b_-^{-1} + L\ell(v_*)) \le e_*^{-\varepsilon/2}.$$

The same upper bound for ℓ_* gives the second and fourth inequalities of (43) from the corresponding polynomial bounds in the inventory; its third inequality is the inventory's charge bound. For the two tails, the supplied estimates give

$$12\ell_*^3 (e^{-p_{0,*}/2} + e^{-p_*^2/(2C_0^2)}) \le 12\ell(v_*)^3 (e^{-v_*^9/2} + e^{-v_*^{38}/(2C_0^2)}) \le e_*,$$

$$C_\Sigma e^{-\gamma_\Sigma \ell_*} \le C_\Sigma e^{-\gamma_\Sigma M v_*^2} \le e_*.$$

This verification applies to every $R \in \mathcal{T}(e)$ above its mesh threshold. The same inventory at v_K gives $e_K^{1/4-9\varepsilon} \le 1/8$. All these conditions are uniform in the cutoff.

Put

$$\varepsilon_1 = \tfrac{1}{2} e_*^{1/4-9\varepsilon}, \qquad \varepsilon_2 = C_\Sigma \ell_*^{-1} e^{-\gamma_\Sigma \ell_*}. \tag{44}$$

Together with (22), (43) gives $\varepsilon_1 \le 1/16$, $\varepsilon_2 \le e_*$, and $\varepsilon_4 \le \varepsilon_3 \le e_*/2$.

Proposition 5.1. For a controlled real pair, every mean-zero real coarse field A with $\|dA\|_\infty \le p_*$ and every $\theta \in \mathbb{R}^3$ satisfy

$$Y_*(A + c_\theta) = e^{-\varepsilon_k^{\mathrm{v}} \ell_*^3} R_\theta^{\mathrm{tw}} (1 + r_*(\theta, A)), \qquad |r_*(\theta, A)| \le \varepsilon_1. \tag{45}$$

The remainder is continuous in (θ, A) on this set. ∎

Proof. The gauge construction in [26d, (32)–(33)] gives a real λ such that $A^g = A - d\lambda$ satisfies $\|A^g\|_\infty \le 2\ell_*^2 p_*$. The curvature bound puts both $A + c_\theta$ and $A^g + c_\theta$ in Dom_k, by [26d, Theorem B]; theorem B gives equality of their terminal densities. Set $\mathscr{A}^0 = H_k^{\mathrm{ph}} A^g$. Summing each of the three photon kernel bounds gives a norm at most $C_0 \|A^g\|_\infty$; hence

$$\|\mathscr{A}^0\|_{\mathrm{each}} \le 2C_0 \ell_*^2 p_* < \tfrac{1}{2} e_*^{-\varepsilon}.$$

The photon minimizer preserves constants, so the background is $\mathscr{A} = \mathscr{A}^0 + c_\theta$. For $\rho_* = e_*^{-3/4+4\varepsilon} \ge 8$ and $|t| < 2\rho_*$, each norm of $t\mathscr{A}^0$ is less than $e_*^{-3/4+3\varepsilon}$, the smallest analytic-domain threshold. Invariance under real constant shifts gives

$$\mathscr{A}^{(t)} = t\mathscr{A}^0 + c_\theta \in \widetilde{R}_k \subset \widetilde{G}_k \qquad (|t| < 2\rho_*). \tag{46}$$

Let $D^{\mathrm{tw}} = D_k(c_\theta)$ and $C_\theta = (D^{\mathrm{tw}})^{-1}$. The free block formula and its twist conjugacy give $\|C_\theta\|_{\mathrm{op}} \le b_-^{-1} + \mu_*^{-1}$. At the connection $\mathscr{A}$, after the fermion field substitution, put

$$\begin{aligned} V_* = & -\langle \bar{\Psi}, [D_k(\mathscr{A}) - D_k(c_\theta)]\Psi \rangle - m_k \langle \bar{\psi}, \psi \rangle \\ & + \sum_X E_k(X; \mathscr{A}, \psi, \chi). \end{aligned} \tag{47}$$

All terms are even. Completing the Grassmann Gaussian in (16), at zero external slot, gives

$$Y_*(A + c_\theta) = e^{-\varepsilon_k^{\mathrm{v}} \ell_*^3} R_\theta^{\mathrm{tw}} \int \mathrm{d}\mu_{C_\theta}(\Psi) e^{V_*}. \tag{48}$$

This is the weight-one identity of the terminal coarse space; the fine fields retain their $\eta_k^3/48$ pairing in V_*.

We estimate the three terms before integrating. Let $a_1 = e_*^{-1/4}$ and $a_2 = C_0^{-1} e_*^{-1/4+\varepsilon} \le a_1$. The analytic quadratic form $f(t) = \langle \bar{\Psi}, D_k(\mathscr{A}^{(t)})\Psi \rangle$ has $|f(t)|_{a_1} \le C_0 e_*^{-1/2} \ell_*^3$ on (46), by (65). Coefficientwise Cauchy's formula with kernel $[t(t-1)]^{-1}$ on $|t| = \rho_*$ gives

$$|f(1) - f(0)|_{a_2} \le |f(1) - f(0)|_{a_1} \le \frac{C_0 e_*^{-1/2} \ell_*^3}{\rho_* - 1} \le 2C_0 e_*^{1/4-4\varepsilon} \ell_*^3.$$

The mass-form and activity-substitution bounds (64) and (63) give

$$|m_k \langle \bar{\psi}, \psi \rangle|_{a_2} \le 4 e_*^{1/4-8\varepsilon} \ell_*^3, \qquad |E_k(X; \mathscr{A}, \psi, \chi)|_{a_2} \le e^{-\kappa d_M(X)} w_k^{\mathrm{E}}.$$

Here $|m_k| \le e_*^{3/4-8\varepsilon}$ and $w_k^{\mathrm{E}} = e_*^{1/4-7\varepsilon}$. These norm estimates include the ordered pairs within one chamber cluster. The mass pairing uses the fine site measure $\eta_k^3/48$; the four spin components in the mass form therefore contribute $4\ell_*^3$, rather than $192\ell_*^3$. Every polymer contains an M-cube, so $\sum_X e^{-\kappa d_M(X)} \le (\ell_*/M)^3 c_\kappa \le c_\kappa \ell_*^3$. Consequently

$$|V_*|_{a_2} \le (2C_0 + c_\kappa + 4) e_*^{1/4-8\varepsilon} \ell_*^3. \tag{49}$$

The covariance integral is the unit-covariance integral after the one-sided substitution $\Psi \mapsto C_\theta \Psi$, $\bar\Psi \mapsto \bar\Psi$. Each covariance row has $4\ell_*^3$ entries of modulus at most $b_-^{-1} + \mu_*^{-1}$. Its row norm, together with the identity on barred variables, is bounded by

$$4\ell_*^3(b_-^{-1} + \mu_*^{-1}) \le \frac{e_*^{-\varepsilon/2}}{2C_0^2\ell_*^3} = a_2 \frac{e_*^{1/4-3\varepsilon/2}}{2C_0\ell_*^3} \le a_2.$$

The weighted substitution estimate therefore sends V_* to V_*' with $|V_*'|_1 \le |V_*|_{a_2}$. The second inequality of (43) and (49) imply $|V_*'|_1 \le e_*^{1/4-9\varepsilon}/4 \le 1/32$. The unit-covariance Wick integral is bounded by the weight-one Grassmann norm and integrates 1 to 1. Thus

$$\left| \int \mathrm{d}\mu_{C_\theta} e^{V_*} - 1 \right| \le |e^{V_*'} - 1|_1 \le e^{|V_*'|_1} - 1 \le 2|V_*'|_1 \le \varepsilon_1.$$

Together with (48), this proves (45). The density is jointly continuous by theorem B. The twist determinant is a continuous positive function, so its quotient in (45) is continuous on the asserted set. □

5.1. **Partition bounds and comparison of torus sizes.** Let $z_{r,n}$ be the matter factor in (1), formed at the real mass $\bar m\eta + m_{\mathrm{in}}$. The fine normalization is

$$z_{r,n} = e^{\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3} \det(D^{\mathrm{ch}}[0] + \bar m\eta)\, \bar\nu(1) > 0. \tag{50}$$

Both signs of the total lattice mass are allowed here. The positive free reference mass $\bar m\eta$ is used only in the determinant normalization.

Proposition 5.2. For every controlled real pair on R, the following two-sided estimates hold:

$$e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3} R_\theta^{\mathrm{tw}}(1-\varepsilon_1)(1-\varepsilon_4) \le \nu^\theta(\mathfrak{G}) \le e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3} R_\theta^{\mathrm{tw}}(1+\varepsilon_1), \tag{51}$$

$$(1-\varepsilon_3)\bar\nu(1) \le \bar\nu(\mathfrak{G}) \le \bar\nu(1). \tag{52}$$

In particular

$$\left|\log z_{r,n} - \log z_{r,n}^{\mathrm{free}} - \varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3 + \varepsilon_k^{\mathrm{v}}\ell_*^3\right| \le 2\varepsilon_1 + \varepsilon_2 + 2\varepsilon_3 + 2\varepsilon_4 \le 2e_*^{1/4-9\varepsilon}, \tag{53}$$

$$\left|\log z_{r,n} - [p_{\mathrm{ch}}^0(\bar m\eta) + \varepsilon_{\mathrm{in}}^{\mathrm{v}}]n^3 + \varepsilon_k^{\mathrm{v}}\ell_*^3\right| \le \varepsilon_5(\bar m R) + 2e_*^{1/4-9\varepsilon}. \tag{54}$$

■

Proof. Apply (21) at $a = 0$ and $\Phi = 1$, then insert (45). The result is

$$\nu^\theta(\mathfrak{G}) = e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3} R_\theta^{\mathrm{tw}} \int \chi_*(A)(1 + r_*(\theta, A))\, \mathrm{d}\mathsf{m}_*(A).$$

The left side and the prefactor are real, and the prefactor is positive. The last integral therefore equals its real part. The remainder bound and $1 - \varepsilon_4 \le \mathsf{m}_*(\chi_*) \le 1$ give (51). The first inequality of (23) gives $\bar\nu(\mathfrak{G}^c) \le \varepsilon_3\bar\nu(1)$. This and positivity prove (52).

Multiply (51) by $\det(D^{\mathrm{ch}}[0] + \bar m\eta) > 0$. The determinant identity (38) and the penalty twist bound replace this product by the fine twisted free determinant, with factors between $e^{-\varepsilon_2}$ and e^{ε_2}. Integrating in θ gives

$$\begin{aligned} &e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3} z_{r,n}^{\mathrm{free}} e^{-\varepsilon_2}(1-\varepsilon_1)(1-\varepsilon_4) \\ &\qquad \le \det(D^{\mathrm{ch}}[0] + \bar m\eta)\bar\nu(\mathfrak{G}) \le e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3} z_{r,n}^{\mathrm{free}} e^{\varepsilon_2}(1+\varepsilon_1). \end{aligned}$$

All integrands are continuous in θ, and the factors being compared are positive. Use (52) and (50), then take logarithms. Writing the expression on the left of (53) without absolute values as Q_n, we obtain

$$-\varepsilon_2 + \log(1-\varepsilon_1) + \log(1-\varepsilon_4) \le Q_n \le \varepsilon_2 + \log(1+\varepsilon_1) - \log(1-\varepsilon_3).$$

The errors lie in $[0,1/2]$. The inequalities $-\log(1-x)\le 2x$ and $\log(1+x)\le x$ therefore give the first bound in (53). Its right side is at most $e_*^{1/4-9\varepsilon}+3e_*$. Since $e_*^{1/4}\le 1/8$, we have $3e_*\le e_*^{1/4}\le e_*^{1/4-9\varepsilon}$, which gives the second bound. Adding (28) proves (54). □

5.2. **Proof of the common-pressure theorem.**

Proof of Theorem A. The common pair is controlled on every torus in the asserted range, by [26a, Theorem D]. Its terminal vacuum obeys $|\varepsilon_k^{\mathrm{v}}|\le 5c_{\mathrm{fl}}e_*^{1/4-6\varepsilon}$. The second inequality of (43) therefore gives

$$|\varepsilon_k^{\mathrm{v}}|\ell_*^3\le \tfrac14 e_*^{1/4-7\varepsilon}\le e_*^{1/4-9\varepsilon}.$$

Remove this term from (53) and (54). This proves the first two estimates in (4). The function ε_5 is decreasing. The scheme's scale bounds give $\bar m R\ge \mu_K>1/L$ and $e_*\le e_K$. Since $1/4-9\varepsilon>0$, they prove its last estimate.

Both R_* and $LR_*=R_{\mathrm{top}}$ belong to $\mathcal{T}(e)$. The mesh threshold is monotone: $N_{\min}(LR_*,e)\le N_{\min}(R_*,e)=N_{\mathrm{ref}}(e)$. Thus the common pair and pressure are available on both tori at every cutoff in the theorem. The bulk pressure cancels exactly:

$$D_{\mathrm{mat}}=L^3(\log z_{r,n_*}-p_Fn_*^3)-(\log z_{r,Ln_*}-p_F(Ln_*)^3).$$

The pressure bounds give the first inequality in (5). Monotonicity of ε_5 and $e_*\le e_K$ give the second. Finally $\bar mR_*>1/L$ and $e_K^{1/4-9\varepsilon}\le 1/8$ show that the bracket is at most $\varepsilon_5(1/L)+1$. The terminal scales, R_* and e_K, are independent of N; this proves the stated uniformity. □

6. Terminal fermion estimates

We prove the operator and coefficient estimates used in the terminal remainder. The reference mass remains variable throughout this section; the interaction mass is part of the Grassmann polynomial in (47). The block operators and their field domains are those of [26b, Theorem A and §4.2] and [26d, §6].

At the stop level write $\eta_*=L^{-k_*}$, $T_*=\eta_*\mathbb{Z}^3/\ell_*\mathbb{Z}^3$ and $T_*^0=\mathbb{Z}^3/\ell_*\mathbb{Z}^3$. The fine chamber spinors have counting measure $\eta_*^3\,48^{-1}\sum_{c,g}$, and the coarse spinors have four components per unit site and weight one. We use

$$k_*\ge 1,\qquad \ell_*>2M+2,\qquad \gamma^FM\ge 16,$$
$$1-L^{-1}=:b_-\le b_*\le 1,\qquad 0<\mu_*\le 1.$$

The running mass μ_* is positive; the bare mass coupling is treated separately. Let

$$F_*(A)=D_A+\mu_*+b_*JP_*(A)J^*,\qquad S_*(A)=F_*(A)^{-1},$$
$$H_*(A)=b_*S_*(A)JQ_*^T(-A),\qquad D_*(A)=b_*-b_*Q_*(A)J^*H_*(A).\tag{55}$$

These are the blocked operators already constructed in [26b, (51)] and [26b, (52)], at $k=k_*$. The symbol F_* here is an operator on spinors, distinct from the curvature variable in the external partition function.

For a unit site y let Δ_y be its fine block. The averaging phases have the form $\varphi_A(y,c)=e^{i\phi_A(y,c)}$, where $\phi_A=e_*\eta_*\tau_{k_*}A$ is linear in A. The nested block centers give

$$|\phi_B(y,c)|\le \tfrac32 e_*|B|_\infty,\qquad |\upsilon_B(c,c')|\le 3|c-c'|e_*|B|_\infty.\tag{56}$$

The first phase uses only bonds inside Δ_y; the second, which defines $U_A(c,c')=e^{i\upsilon_A(c,c')}$, uses only bonds in the coordinate box between its endpoints. For the first bound, the path lengths at block scales $L^j\eta_*$ add up to $\frac32(L-1)\eta_*\sum_{j<k_*}L^j\le\frac32$. Each of the six coordinate-order paths between c and c' has length at most $3|c-c'|$, which gives the second bound.

Fix the exponent $1-\varepsilon<\alpha<1$ from the construction. For distinct chamber sites $x=(c,g)$ and $x'=(c',g')$, the pair convention of [26g, §2.2] is

$$d_X(x,x')=\max\{|c-c'|_\infty,\eta_*\}\le M,\qquad (\delta_{\alpha,A}v)(x,x')=\frac{U_A(c,c')v(x')-v(x)}{d_X(x,x')^\alpha}.$$

Distances use the shortest simultaneous periodic lift. For pairs within one cluster, $d_X=\eta_*$ and $U_A=1$. The barred quotient uses U_A^{-1}. The admissible pairs for each polymer, including its boundary pairs of chamber bond type 3, are as in that convention.

For a complex direction B, let P_B be the unit sites y whose block Δ_y contains an endpoint of a bond in $\operatorname{supp}B$. Define

$$\epsilon_B=e_*|B|_\infty,\qquad P_B^+=\{y:d(y,P_B)\le M+1\}.$$

The distance is the periodic sup distance on the unit lattice. For an operator O between fine, coarse or pair spaces, localize each fine spinor in its unit block and each pair at its first point, and put

$$|O|(y,y')=\sup_{\|v\|_\infty\le1}\|\mathbf{1}_yO\mathbf{1}_{y'}v\|_\infty,$$
$$\|O\|_\gamma=\max\left\{\sup_y\sum_{y'}|O|(y,y')e^{\gamma d(y,y')},\sup_{y'}\sum_y|O|(y,y')e^{\gamma d(y,y')}\right\}.\tag{57}$$

The spinor sup norm uses the maximum of component moduli. Inserting the unit-block partition between two factors gives $|O_1O_2|\le|O_1||O_2|$ and $\|O_1O_2\|_\gamma\le\|O_1\|_\gamma\|O_2\|_\gamma$. For coarse spinor operators,

$$\|O\|_{\ell^2\to\ell^2}\le2\|O\|_0,\qquad \|O\|_{\mathfrak{S}_1}\le4\sum_{y,y'}|O|(y,y').\tag{58}$$

Indeed the matrix row sums are bounded by $\|O\|_0$ and the column sums by $4\|O\|_0$, so Schur's test gives the first bound. Polar decomposition bounds the trace norm by the sum of all entry moduli, and the entries in one block have total modulus at most four times its maximum row sum.

We use the fixed constants C^F,γ^F of the fermion construction, with its calibration $a'\ge1$ and

$$C^F\ge608\exp(1)\max(1,b)a',\qquad C^F\ge64\exp(1)b^2a',\qquad b=1.$$

The first inequality also dominates the component norm bounds $64a'$ and $608a'$ for the fine inverse and its short covariant quotients. The charge ceiling satisfies $6Le_*^\varepsilon\le1$ and $3Me_*^\varepsilon\le1$; on $\widetilde G_*$, $\mathfrak i(A)<e_*^\varepsilon$. The phase estimates therefore give $|\varphi_A^{\pm1}|,|U_A^{\pm1}|\le\exp(1)$, including every allowed long pair. The unweakened estimate [26b, (43)], at its spectral parameter $y=\infty$, and the block phases give the following terminal bounds. On the outer field domain $\widetilde G_*$ the inverse S_* is analytic, and, with $E(y,y')=e^{-\gamma^Fd(y,y')}$,

$$|S_*|,\quad|\delta_{\alpha,A}S_*|,\quad|H_*|,\quad|\delta_{\alpha,A}H_*|\le C^FE.\tag{59}$$

The two pair majorants are restricted to $d_X<1$, including pairs inside a chamber cluster. The field domain

$$\mathcal W=\{A_r+\mathscr C:A_r\in\widetilde R_*\text{ real},\ \mathscr C\in\mathfrak X_*\}$$

is contained in $\widetilde G_*$, is invariant under $A\mapsto-A$, and satisfies $\mathfrak i(A):=e_*|\operatorname{Im}A|_\infty<1/16$ and $3M\mathfrak i(A)<1/4$. For long pairs, $|\delta_{\alpha,A}v(x,x')|_c\le\mathrm e|v(x')|_c+|v(x)|_c$. The constants $C^F\ge1$ and $0<\gamma^F\le1$ are uniform in the mesh, allowed torus and running mass. These are the stop-level forms of the walk and local inverse estimates of [26b, §3], uniform for $0<\mu_*\le1$. To obtain the bounds for H_*, write $H_*=b_*S_*JQ_*^T(-A)$. The last factor has Euclidean spinor sup norm at most $\exp(1)$, preserves each unit block, and has no chamber factor: it repeats the same coarse spinor over the 48 labels, multiplied by the block phase. Composition with the localized inverse bounds gives constants $32\exp(1)ba'$ and $304\exp(1)ba'$. Since $|v|_c\le|v|_2\le2|v|_c$ on $\mathbb C^4$, their component sup norms are at most twice these constants, hence at most C^F.

We shall also need the global component row bound, without an exponential weight:

$$|H_*^{\#}(A)|_{1,\infty} \le C^F, \qquad |\bar{H}_*^{\#}(A)|_{1,\infty} \le C^F. \tag{60}$$

Here the rows include every allowed pair, and the columns are the four coarse components at the unit sites. Indeed, the unweighted consequences of [26b, (43)] give

$$\|H_*(A)v\|_{\mathrm{su}} \le 32\exp(1)ba'\|v\|_{\mathrm{su}}, \qquad \|\delta_{\alpha,A}H_*(A)v\|_{\mathrm{su}} \le 304\exp(1)ba'\|v\|_{\mathrm{su}}$$

for short pairs. These include distinct labels at the same photon site. For a long pair, $1 \le d_X(x,x') \le M$, so $d_X^{-\alpha} \le 1$ and $|U_A| \le \exp(1)$. Consequently

$$|(\delta_{\alpha,A}H_*(A)v)(x,x')|_2 \le (1+\exp(1))\|H_*(A)v\|_{\mathrm{su}} \le 128\exp(1)ba'\|v\|_{\mathrm{su}}.$$

Thus $304\exp(1)ba'$ bounds the Euclidean sup norm for all pair rows. Conversion to the component sup norm costs a factor of two, and the calibration of C^F gives (60). To identify this operator norm with the row sum, fix one output component and prescribe each input component to have the conjugate phase of its matrix entry. The coarse spinor values on a fundamental domain can be prescribed independently; their antiperiodic extension does not affect the entry moduli. All unit sites have weight one.

The barred bound follows from $\bar{H}_*(A) = \mathsf{C}^{-1}H_*(-A)\mathsf{C}$. The charge-conjugation matrix C is a signed permutation of the four components, and commutes with the scalar phases in the covariant quotient. It preserves both spinor norms. Applying the preceding argument at $-A$ proves the second inequality in (60). Finally, localization of the long-pair argument gives the bound $C^F\exp(-\gamma^F\min\{d([x],y'),d([x'],y')\})$ for the coefficient at a coarse site y'. The intermediate Euclidean constant is $128\exp(1)ba'$, and its component constant $256\exp(1)ba'$ is again at most C^F.

6.1. **The real covariance.** The covariance at a real field requires no small-field hypothesis. If A_r is real, $R = D_{A_r} + \mu_*$ has real part at least μ_*. The averaging map $Q = Q_*(A_r)J^*$ satisfies $Q^* = JQ_*^T(-A_r)$ and $QQ^* = 1$, because its phases have modulus one and every block has fine measure one. Thus $F_*(A_r) = R + b_*Q^*Q$ also has real part at least μ_*. Set $C = QR^{-1}Q^*$ and $Y = (b_*^{-1}+C)^{-1}$. The inverse R^{-1} is accretive, so $b_*^{-1}+C$ is invertible. Multiplication verifies

$$S_* = R^{-1} - R^{-1}Q^*YQR^{-1}, \qquad QS_*Q^* = b_*^{-1}(1-b_*^{-1}Y).$$

The definition of D_* therefore gives

$$D_*(A_r)^{-1} = b_*^{-1} + QR^{-1}Q^*, \qquad \|D_*(A_r)^{-1}\|_2 \le b_*^{-1} + \mu_*^{-1}. \tag{61}$$

Its block decay follows by conjugating the fine inverse with a weight. Fix y' and put $\psi(c) = \vartheta\, d(c,\Delta_{y'})$ on the periodic fine lattice, with $\vartheta = \mu_*/(12\mathrm{e}) \le 1$. This weight is zero on $\Delta_{y'}$, is ϑ-Lipschitz, and on Δ_y is at least $\vartheta(d(y,y')-1)$. Only the external Dirac hop changes under conjugation by e^{ψ}, since the internal hops stay over one photon site. Consequently

$$\|e^{\psi}D_{A_r}e^{-\psi} - D_{A_r}\|_2 \le 6\eta_*^{-1}(e^{\vartheta\eta_*}-1) \le 6\mathrm{e}\vartheta = \mu_*/2.$$

The conjugated R has real part at least $\mu_*/2$, so its inverse has norm at most $2/\mu_*$. Apply it between Q^* times unit spinors at y and y'; these have norm one and supports in their respective blocks. Moving the weight across the scalar product gives

$$\|D_*(A_r)^{-1}(y,y')\|_2 \le b_*^{-1}1_{\{y=y'\}} + \frac{2\mathrm{e}}{\mu_*}e^{-\mu_* d(y,y')/(12\mathrm{e})}. \tag{62}$$

The factor e accounts for the unit block diameter in the lower bound for ψ.

6.2. **The terminal Grassmann norms.** The terminal integral is over four coarse components per unit site, with Grassmann weight one. For a polynomial $F = \sum_I F_I \Psi_I$ in ordered distinct generators, write $|F|_a = \sum_I |F_I| a^{|I|}$. The fine norm $\|F\|_{h_*}$ uses the prescribed fine measure and the two weights for spinor and Hölder-pair generators, as in [26g, (4)–(5)]; the weighted substitution estimate is proved in [26b, Proposition 5.1]. Set

$$a_1 = e_*^{-1/4}, \qquad a_2 = C_0^{-1} e_*^{-1/4+\varepsilon}, \qquad h_* = (e_*^{-1/4}, e_*^{-1/4+\varepsilon}).$$

By (60), the substitution $H_*^{\#}(A)\Psi$ has row norm at most $C^F \le C_0$, for both spinor and Hölder legs, including all long and cluster pairs. The coarse kernel $b_* - D_*(A) = b_*^2 Q_* J^* S_* J Q_*^T(-A)$ is bounded by $C^F E(y,y')$: block compression gives Euclidean constant $32\exp(1) b^2 a'$ by [26b, (60)], and conversion to the component norm doubles it. The calibration of C^F applies.

The weighted Grassmann substitution estimate applies with $C_0 a_2 = h_{*,2} \le h_{*,1}$. Therefore

$$|F(H_*^{\#}(A)\Psi)|_{a_2} \le \|F\|_{h_*}. \tag{63}$$

For the mass form before substitution, the fine measure sums to ℓ_*^3 and there are four spin components. Its norm at weight $C_0 a_2$ is exactly $4\ell_*^3 (C_0 a_2)^2$. In particular

$$|\langle \bar\psi, \psi\rangle|_{a_2} \le 4 e_*^{-1/2+2\varepsilon} \ell_*^3. \tag{64}$$

The 48 labels are canceled by the fine measure $1/48$; there is no factor 192 in this estimate. Similarly the coarse bilinear form has norm a_1^2 times the sum of its matrix entry moduli. Its diagonal term contributes $4b_*$ per unit site, and its decaying kernel contributes at most $4C^F \sum_{y'} E(y,y')$. For $c(\gamma) = \sum_{z\in\mathbb{Z}^3} e^{-\gamma|z|_\infty}$, the lattice-sum estimate in appendix A gives $c(\gamma) \le (12/\gamma)^3$ when $0 < \gamma \le 6$. The chosen $C_0 \ge 4 + 16 C^F c(\gamma^F)$ therefore gives

$$|\langle \bar\Psi, D_*(A)\Psi\rangle|_{a_1} \le C_0 e_*^{-1/2} \ell_*^3. \tag{65}$$

These estimates hold on $\widetilde{G}_*$. For the external activity evaluated at $\mathscr{T} = \mathscr{A} + \mathscr{C}$, the same substitution applies because $\widetilde{R}_* + \mathfrak{X}_* \subset \widetilde{G}_*$.

Appendix A. Marked insertions and reference-mass covariance

The terminal operators of section 6 also give estimates for local variations of the connection. These estimates are used for the differentiated terminal integral. The first subsection uses the field domain $\mathcal{W}$ and the operator bounds already stated. For the neighborhood and covariance constants in the second subsection we additionally fix

$$L^{-3} =: \mu_- < \mu_* \le L^{-1}.$$

This interval holds at the reference stop; it is not imposed in theorem A.

A.1. **Local derivatives of the block operator.** Put

$$\mathcal{S} = C^F (24/\gamma^F)^3, \qquad h_{\mathrm{L}} = 6\mathcal{S}, \qquad C_H = 203 M (C^F)^2, \qquad C_K = 226 \mathcal{S}^2.$$

The lattice sum $\sum_{y'} e^{-\gamma d(y,y')} \le (12/\gamma)^3$, $0 < \gamma \le 6$, follows by choosing nearest periodic representatives and summing $e^{-\gamma|z|_\infty} \le e^{-\gamma|z|_1/3}$ over $\mathbb{Z}^3$. The one-dimensional sum is $\coth(\gamma/6) \le 12/\gamma$.

For a fine spinor or Hölder-pair index u, let $\mathrm{Leg}_A(u; y', \beta')$ be the coefficient of $\Psi_{\beta'}(y')$ after substituting $H_*^{\#}(A)\Psi$ in that generator. Barred generators use the corresponding barred coefficient. The set $\mathrm{pos}(u)$ consists of the unit block containing the first endpoint, and, for a pair with $d_X(x,x') \ge 1$, also the block containing its second endpoint. Thus it has at most two elements.

Proposition A.1. For $A \in \mathcal{W}$, the blocked field maps and their Hölder maps are analytic in A. The first coarse-operator derivative satisfies

$$|\partial_B D_*(A)|(y,y') \le 226(C^F)^2\epsilon_B \sum_{z\in P_B} E(y,z)E(z,y'),$$
$$\|\partial_B D_*(A)\|_{\gamma^F/2} \le C_K\epsilon_B, \qquad \sum_{y,y'} |\partial_B D_*(A)|(y,y') \le C_K\epsilon_B|P_B|. \tag{66}$$

If $d(P_B,P_{B'}) \ge M+2$, then

$$|\partial_B\partial_{B'} D_*(A)|(y,y') \le C_H^2\epsilon_B\epsilon_{B'} \sum_{z\in P_B, z'\in P_{B'}} (E(y,z)E(z,z')E(z',y') + E(y,z')E(z',z)E(z,y')). \tag{67}$$

The corresponding fine-field legs satisfy

$$|\operatorname{Leg}_A|(u;y') \le 3C^F \sum_{w\in \operatorname{pos}(u)} E(w,y'),$$
$$|\partial_B \operatorname{Leg}_A|(u;y') \le C_H\epsilon_B \sum_{w\in\operatorname{pos}(u), z\in P_B^+} E(w,z)E(z,y'), \tag{68}$$
$$|\partial_B\partial_{B'} \operatorname{Leg}_A|(u;y') \le C_H^2\epsilon_B\epsilon_{B'} \sum_{\substack{w\in\operatorname{pos}(u)\\ z\in P_B^+, z'\in P_{B'}^+}} (E(w,z)E(z,z')E(z',y') + E(w,z')E(z',z)E(z,y')). \tag{69}$$

The last bound has the same separation hypothesis. These estimates hold for barred and unbarred legs. ∎

Proof. The block averages and their transposes are diagonal in unit blocks. From (56) and $\mathfrak{i}(A) \le 1/4$ their majorants are at most $3/2$. Their directional derivatives are bounded by

$$(9/4)\epsilon_B 1_{\{y\in P_B\}}.$$

Differentiating the only phased Dirac hop gives

$$(\partial_B D_A v)(x) = -ie_*B(c,c_3)M_3(x)u_A(c,c_3)v(x^{(3)}).$$

Its majorant is at most $8\epsilon_B$, vanishes unless the unit blocks coincide or are neighbors, and vanishes unless both belong to P_B. The derivative of $JP_*(A)J^*$ has diagonal majorant at most $7\epsilon_B$. Thus for $F_B = \partial_B F_*(A)$,

$$|F_B|(z,z') \le \epsilon_B(15\,1_{\{z=z'\}} + 8\,1_{\{z,z' \text{ neighbors}\}})1_{\{z,z'\in P_B\}}.$$

Summing the six neighboring blocks gives $\|F_B\|_\gamma \le 146\epsilon_B$ for $0 \le \gamma \le 1$. Also, since $E(z',y') \le \mathrm{e}E(z,y')$ for neighboring z,z',

$$\sum_{z,z'} E(y,z)|F_B|(z,z')E(z',y') \le 146\epsilon_B \sum_{z\in P_B} E(y,z)E(z,y'). \tag{70}$$

If the supports are separated by $M+2$, the mixed derivatives of D_A, the averages, JP_*J^*, F_* and $\delta_{\alpha,A}$ vanish. Each entry is an exponential of a linear phase supported in a unit block, a single external bond or a coordinate box of diameter at most M. Two separated directions cannot both occur in one phase. For JP_*J^* the remaining product terms have disjoint unit-block supports.

The inverse derivative and the blocked-field definition give

$$\partial_B H_* = -S_*F_BH_* + b_*S_*\partial_B(JQ_*^T(-A)).$$

Use (59) and (70) in the first term. The diagonal derivative in the second is absorbed as the summand $z = y'$. The combined coefficient is at most $149(C^F)^2$:

$$|\partial_B H_*|(y,y') \le 149(C^F)^2 \epsilon_B \sum_{z\in P_B} E(y,z)E(z,y'). \tag{71}$$

The same estimate holds for $\delta_{\alpha,A}\partial_B H_*$ on short pairs, since the first factor S_* can be replaced by $\delta_{\alpha,A}S_*$.

For separated directions differentiating once more leaves four terms:

$$\begin{aligned}\partial_B\partial_{B'}H_* = {} & S_*F_BS_*F_{B'}H_* + S_*F_{B'}S_*F_BH_* \\ & - b_*S_*F_BS_*\partial_{B'}(JQ_*^T(-A)) - b_*S_*F_{B'}S_*\partial_B(JQ_*^T(-A)).\end{aligned}$$

Apply (70) twice to the first two terms. Each gives an ordered three-link exponential chain with coefficient $146^2(C^F)^3$. The diagonal last factors in the other two terms give the same chains, with coefficient $(657/2)(C^F)^2$. Each ordered chain therefore has coefficient at most $21645(C^F)^3$. Again the first factor may be replaced by the short-pair Hölder inverse.

Differentiate $D_* = b_* - b_*Q_*J^*H_*$. The derivative of the average contributes $(9/4)C^F$ to the first exponential chain and the derivative of H_* contributes $(3/2)149(C^F)^2$. Their sum is at most $226(C^F)^2$, proving the first line of (66). Weighted row and column sums give two factors $(24/\gamma^F)^3$, hence C_K; without weights the same sum retains one factor $|P_B|$. For the mixed derivative the three product terms contain one derivative on each factor, or the mixed derivative of H_*. Each chain has coefficient at most $(1341/4)(C^F)^2 + (64935/2)(C^F)^3 \le 32803(C^F)^3 \le C_H^2$. This proves (67).

A leg is a component row of H_* or of $\delta_{\alpha,A}H_*$. For a short pair its position is its first unit block; for a long pair its position set contains both endpoint blocks. The sum over spin components of the row is bounded by the relevant majorant. For long pairs use $|\delta_{\alpha,A}v| \le \mathrm{e}|v(x')| + |v(x)|$; for short pairs use the Hölder input directly. This gives the first line of (68). In particular, with $d(y', \mathrm{pos}(u)) = \min_{w\in\mathrm{pos}(u)} d(y', w)$,

$$\sum_{y',\beta'} |\operatorname{Leg}_A(u; y', \beta')| e^{(\gamma^F/2)d(y',\mathrm{pos}(u))} \le 3C^F |\operatorname{pos}(u)| (24/\gamma^F)^3 \le h_{\mathrm{L}}.$$

Indeed, multiplying each endpoint decay by this weight leaves decay rate at least $\gamma^F/2$, and there are at most two endpoints.

The derivative of a Hölder leg has two terms: $(\partial_B\delta_{\alpha,A})H_* + \delta_{\alpha,A}\partial_B H_*$. The first is bounded by $4d_X^{1-\alpha}\epsilon_B|H_*v(x')|$, and vanishes unless the pair is within M of P_B. It vanishes for pairs inside a cluster, since their phase is one. For short pairs this adds at most $11C^F$ to $149(C^F)^2$; for long pairs it adds $4MC^F$ to $(4/3)149(C^F)^2$. Both are at most C_H, after enlarging P_B to P_B^+. This proves the derivative bound in (68).

For its mixed derivative the second derivative of the Hölder phase vanishes by separation. There remain $(\partial_B\delta_{\alpha,A})\partial_{B'}H_*$ and its interchange, and $\delta_{\alpha,A}\partial_B\partial_{B'}H_*$. On short pairs each ordered chain has coefficient at most $1639(C^F)^2 + 21645(C^F)^3$. On long pairs the coefficient at the second endpoint is at most $596M(C^F)^2 + 28860(C^F)^3$; the first endpoint has the smaller coefficient $21645(C^F)^3$. All are at most C_H^2, proving (69). Barred legs follow by the monomial charge conjugation matrix and the replacement $A \mapsto -A$, which preserve component row sums; the domain is invariant under that replacement. □

A.2. **A complex neighborhood and covariance bounds.** Choose

$$c_\tau = \min\{1/8, \mu_-/48, (292\mathcal{S})^{-1}\}, \qquad C_K' = 1328\mathcal{S}^2, \qquad \mathcal{W}^+ = \{A + A_1 : A \in \mathcal{W},\ e_*|A_1|_\infty \le c_\tau\}.$$

Every point of $\mathcal{W}^+$ has imaginary part $\mathfrak{i} < 1/4$. The elementary averages and F_* therefore satisfy the same local derivative bounds as above, although the exponential propagator input is initially

given only on $\mathcal{W}$. The enlargement preserves invertibility:

$$\|S_*(A+A_1)\|_{\gamma^F/2} \le 2\mathcal{S}. \tag{72}$$

To prove this, integrate $\partial_{A_1} F_*$ along the segment to get $\|F_*(A+A_1)-F_*(A)\|_{\gamma^F/2} \le 146 c_\tau$. Multiplying by $S_*(A)$ gives norm at most $146 c_\tau \mathcal{S} \le 1/2$. The Neumann inverse in the submultiplicative majorant norm proves (72). Since its finite-dimensional determinant is nonzero, S_* is analytic in a neighborhood of each point of $\mathcal{W}^+$, as is D_*.

Differentiate $D_* = b_* - b_*^2 Q_* J^* S_* J Q_*^T(-A)$. The three terms have majorant-norm coefficients

$$\tfrac{9}{4}(2\mathcal{S})\tfrac{3}{2}, \qquad \tfrac{3}{2}(2\mathcal{S})146(2\mathcal{S})\tfrac{3}{2}, \qquad \tfrac{3}{2}(2\mathcal{S})\tfrac{9}{4}.$$

Their sum is $\frac{27}{2}\mathcal{S} + 1314\mathcal{S}^2 \le C'_K$. Keeping the support indicator of either the average derivative or F_B gives, respectively,

$$\|\partial_B D_*(A)\|_{\gamma^F/2} \le C'_K \epsilon_B, \qquad \sum_{y,y'} |\partial_B D_*(A)|(y,y') \le C'_K \epsilon_B |P_B|, \qquad A \in \mathcal{W}^+. \tag{73}$$

In the middle term, sum first the two outer S_* factors using their row and column bounds $2\mathcal{S}$, and then sum the local derivative over its first endpoint in P_B. Integrating (73) along any segment inside $\mathcal{W}^+$ gives the same two bounds for $D_*(A+B)-D_*(A)$.

Set

$$g = (b_-^{-1}+\mu_-^{-1})^{1/2}, \qquad h_G = 1+g, \qquad g_1 = b_-^{-1} + 4/\mu_-, \qquad \theta_{\mathrm{CT}} = \mu_-/(12\mathrm{e}),$$
$$\vartheta_0 = \min\{\theta_{\mathrm{CT}}/2, \gamma^F/8, 1/M\}, \qquad \alpha_G = 2b_-^{-1} + 4\mathrm{e}\mu_-^{-1}(24/\theta_{\mathrm{CT}})^3.$$

A component row or column of a 4×4 matrix has sum at most twice its Hilbert-space norm. Using (62) and the lattice exponential sum, every row and column of $G = D_*(A_r)^{-1}$ therefore satisfies

$$\sum_{u'} |G(u,u')| e^{\vartheta_0 d(u,u')} \le \alpha_G, \qquad \sum_{u} |G(u,u')| e^{\vartheta_0 d(u,u')} \le \alpha_G, \qquad \|G\|_2 \le g^2. \tag{74}$$

These constants depend on L, C^F, γ^F, M, not on the size of the real connection.

The same inverse identity gives a bound for the coarse covariance at complex fields. Suppose $A = A_r + A_1$, with A_r real, $e_*|A_1|_\infty \le \mu_*/24$, and $F_*(A)$ invertible. The external-hop phase difference gives $\|D_A - D_{A_r}\|_2 \le 12 e_* |A_1|_\infty \le \mu_*/2$. Hence $D_A + \mu_*$ is accretive with inverse norm at most $2/\mu_*$. Its two average maps have norm at most $\sqrt{2}$, because $|\varphi_A^{\pm 1}| \le e^{1/16} \le \sqrt{2}$. The same finite-rank inversion calculation then gives

$$D_*(A)^{-1} = b_*^{-1} + Q_*(A) J^* (D_A+\mu_*)^{-1} J Q_*^T(-A), \qquad \|D_*(A)^{-1}\|_2 \le g_1. \tag{75}$$

Here invertibility of $b_*^{-1} + QR^{-1}Q'$ follows from that of $F_*(A)$ and R: $\det(1 + b_* R^{-1} Q' Q) = \det(1 + b_* Q R^{-1} Q')$. This determinant identity follows by the two block-triangular factorizations of $\left(\begin{smallmatrix} 1 & -b_* R^{-1} Q' \\ Q & 1 \end{smallmatrix}\right)$. The inverse formula is then checked by multiplication exactly as above. Each covariance entry is bounded by g_1, so each row has sum at most $4\ell_*^3 g_1$.

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